Qualcomm Technologies, Inc.

Device description

The SW5100-0/SW5100P-0 is a wearable chip in 4 nm-LPX process having next generation Qualcomm® Snapdragon™ processor SW5100 and Power Management IC PMW5100 on the same 10.4 mm × 8.6 mm package.

- 10.4 mm × 8.6 mm × 0.48 mm package-on-package (PoP)
  - 0.35 mm pitch – bottom
  - 0.4 mm pitch – top (ePoP)
- Arm Cortex-A53 quad-core applications processor at 1.7 GHz
- Qualcomm® Adreno™ 702 graphics processing unit (GPU) 700 MHz nominal, 1010 MHz turbo with 64-bit addressing
- DSP shared between Snapdragon sensor core and low power audio subsystem
- Package-on-package (PoP) high-speed eMCP memory, 2133 MHz LPDDR4X SDRAM and eMMC 5.1

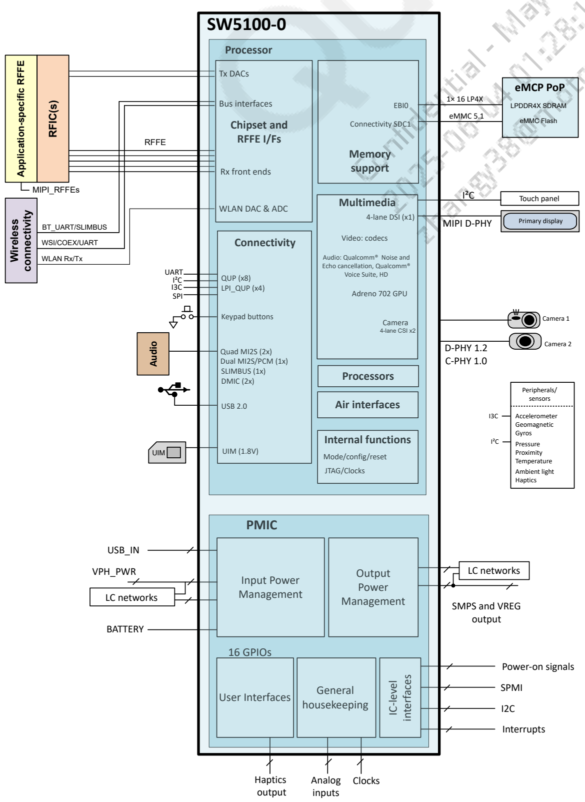
The latest air interface standards are supported only for SW5100-0, including:

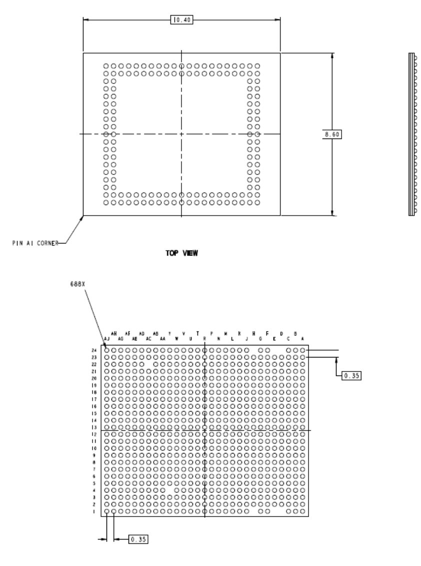
- LTE Cat4
- DC-HSPA+
- WCDMA
- CDMA up to 1X advanced and 1 × EV-DOrA

Key features

- LTE multimode modem paired with Qualcomm Technologies Inc. (QTI) RF transceiver WTR3925:
  - 3GPP Rel. 13 LTE Cat4/Cat1 support (no CA)
  - L1, L5 GNSS support for navigation applications
- 640 × 640 60 fps, 4-layers, 1 × 4-lane DSI
- 16 MP, 1080p, Secure camera, 4 + 4 CSI (D-PHY v1.2) and C-PHY v1.0
- Support for eMMC 5.1, USB 2.0
- QCC5100: WLAN 1 × 1 802.11a/b/g/n, Bluetooth 5.2
- Switched-mode battery charger with boost mode for class D speaker and heart rate monitor
- Qualcomm battery gauge optimized for wearables
- Class D speaker amplifier with two analog MIC inputs
- LRA and ERM haptics support
- Support for external 32.768 kHz clock input, 3x low noise base band clock output, 3x low noise RFCLK clock output, and 3x sleep clock output
- Supports five SMPS, 29 LDO regulators, and one buck-or-boost (BOB) regulator
- 16 GPIOs
- SPMI and I²C interface; I²C interface (co-processor to PMIC)

SW5100-0/SW5100P-0 high-level block diagram and MPSP 688 package outline drawing





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1 Introduction

Document updates

See the Revision history for details on the changes included in this revision.

1.1 Functional block diagram

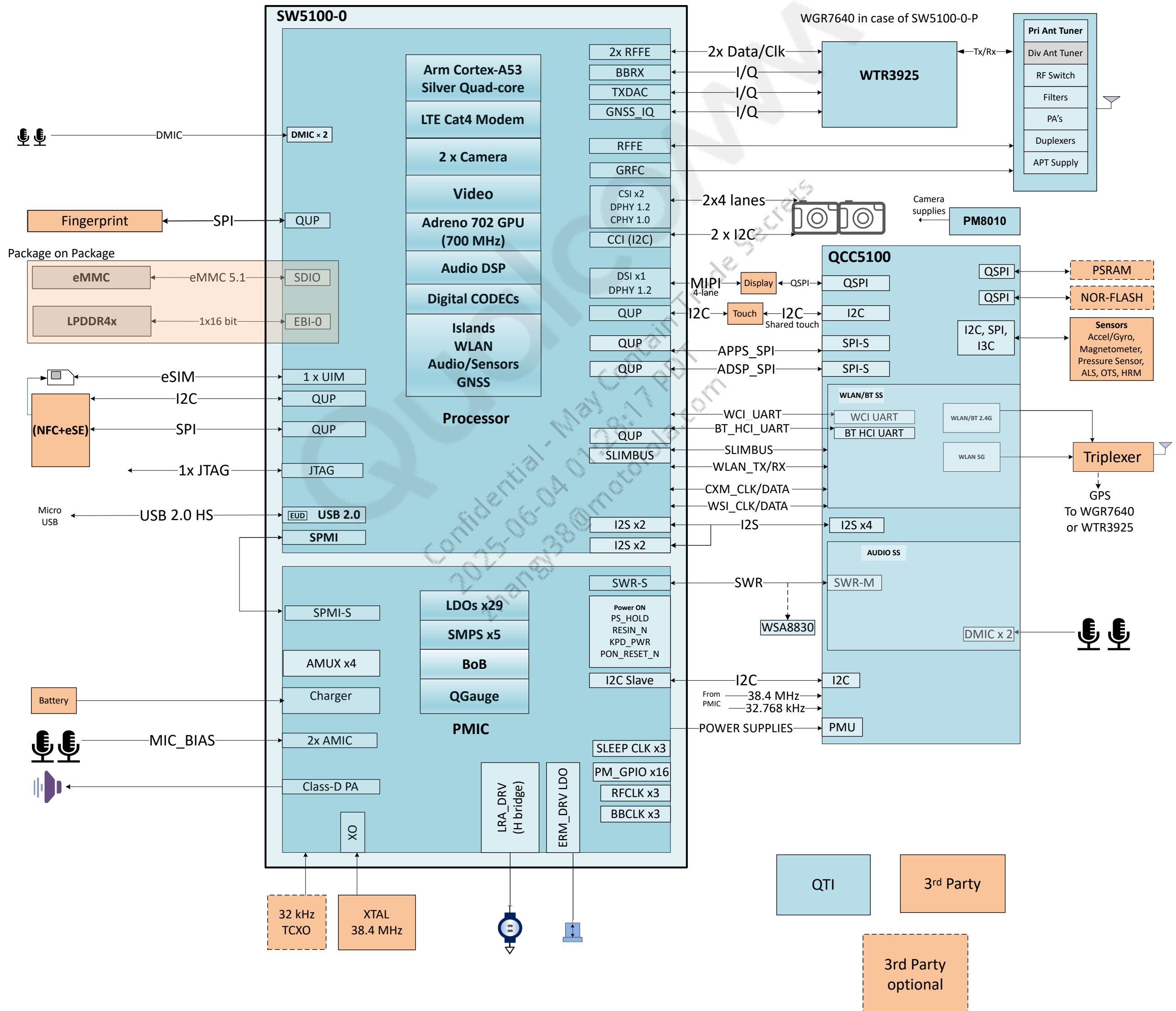

Figure 1-1 SW5100-0/SW5100P-0 functional block diagram and example application

1.2 SW5100-0/SW5100P-0 features

NOTE: Some of the hardware features integrated within the SW5100-0/SW5100P-0 must be enabled by software. See the latest revision of the applicable software release notes to identify the enabled SW5100-0/SW5100P-0 features.

The following table lists the differences between SW5100-0 and SW5100-1 chipsets:

Table 1-1 Comparison of SW5100-0 and SW5100-1 chipsets

Features		SW5100-0	SW5100-1
Dimension		10.4 mm × 8.6 mm × 0.48 mm	8 mm × 9.5 mm × 0.69 mm
Low power co-processor attach		Pairs with QCC5100	No support
PMIC attach		PMW5100 in-built	No in-built PMIC
RF support	RF transceiver	**WTR3925**	**WTR2965**
	RF operating bands	Defined by **WTR3925**	Defined by **WTR2965**
	CA support	Not supported	Not supported
	3GPP Release	Rel.13 support, LTE Cat4/Cat1	Rel.13 support, LTE Cat4/Cat1
WLAN/BT	Chip attach	**QCC5100**	**WCN3980**
	WLAN version	802.11a/b/g/n HT 1 × 1 20 MHz	802.11a/b/g/n/ac VHT 1 × 1 40 MHz
	Bluetooth version	Bluetooth 5.2	Bluetooth 5.1
GNSS	Chip attach	WGR7640 or **WTR3925**	WGR7640 or **WTR2965**
	Bands	GNSS L1 and **L5** bands	GNSS L1 band

NOTE Bold text indicates the differences between SW5100-0 and SW5100-1 chipsets.

For complete list of SW5100 processor features, see *SW5100-1/SW5100P-1 Data Sheet* (80-24634-1).

For complete list of PMW5100 PMIC features, see *PMW5100 Data Sheet* (80-22756-1).

2 Pin definitions

2.1 I/O parameter definitions

Table 2-1 I/O description (Pin type) parameters

Symbol	Description
Pin attribute	
AI	Analog input (does not include pin circuitry)
AO	Analog output (does not include pin circuitry)
BI	Bidirectional digital with CMOS input
DI	Digital input (CMOS)
DO	Digital output (CMOS)
H	High-voltage tolerant
S	Schmitt trigger input
Z	High-impedance (Hi-Z) output
Pin pull details for digital I/Os	
nppdpukp	Programmable pull resistor. The default pull direction is indicated using capital letters, and is a prefix to other programmable options: NP: pdpukp = default no-pull, with programmable options following the colon (:) PD: nppukp = default pull-down, with programmable options following the colon (:) PU: nppdkp = default pull-up, with programmable options following the colon (:) KP: nppdpu = default keeper, with programmable options following the colon (:)
KP	Contains an internal weak keeper device (keepers cannot drive external buses)
NP	Contains no internal pull
PU	Contains an internal pull-up device
PD	Contains an internal pull-down device
Pin-voltage groupings	
P0	Pin group 0; tied to VDD_PX0 pins (1.8 V only)
P1	Pin group 1 (EBI0); tied to VDD_PX1 pins (1.2 V only)
P3	Pin group 3 (most peripherals); tied to VDD_PX3 pins (1.8 V only)
P5	Pin group 5 (UIM1); tied to VDD_PX5 pins (1.8 V or 2.95 V)
P11	Pin group 11; tied to VDD_PX11 pins (1.8 V)
CSI	Supply voltage for MIPI_CSI circuits and I/Os; tied to VDD_A_CSI_1P2 (1.2 V)
DSI	Supply voltage for MIPI_DSI circuits and I/Os; tied to VDD_A_DSI_1P2 (1.2 V)
GNDC	Common ground; a pin that does not handle a significant amount of current flow, typically used for grounding digital circuits and substrates
GNDP	Power ground; a pin that handles 10 mA or more of current flow returning to ground. Layout considerations must be made for these pins.

2.2 Pin assignments – bottom

2.2.1 Pin map – bottom

The SW5100-0/SW5100P-0 is available in the MPSP 688. Its bottom surface is equivalent to a MPSP 688 that includes several ground pins for electrical grounding, mechanical strength, and thermal continuity.

A high-level view of the bottom pin assignments is shown in the following figure.

The text within Figure 2-1 is difficult to read when viewing an 8½ inch × 11 inch hard copy. Other viewing options are available:

- Print that one page on an 11 inch × 17 inch sheet.
- View the graphic's PDF soft copy and zoom in — the resolution is sufficient for comfortable reading.
- Download the *SW5100-0/SW5100P-0 Pin Assignment and GPIO Configuration Spreadsheet* (80-24633-1A). This Microsoft Excel spreadsheet lists all SW5100-0/SW5100P-0 GPIOs, pin numbers, pin names, pin voltages, pin types, and functional descriptions.

NOTE Click the following link to download the pin assignment and GPIO configuration spreadsheet from the CreatePoint website.

https://createpoint.qti.qualcomm.com/search/contentdocument/stream/dcn/80-24633-1A

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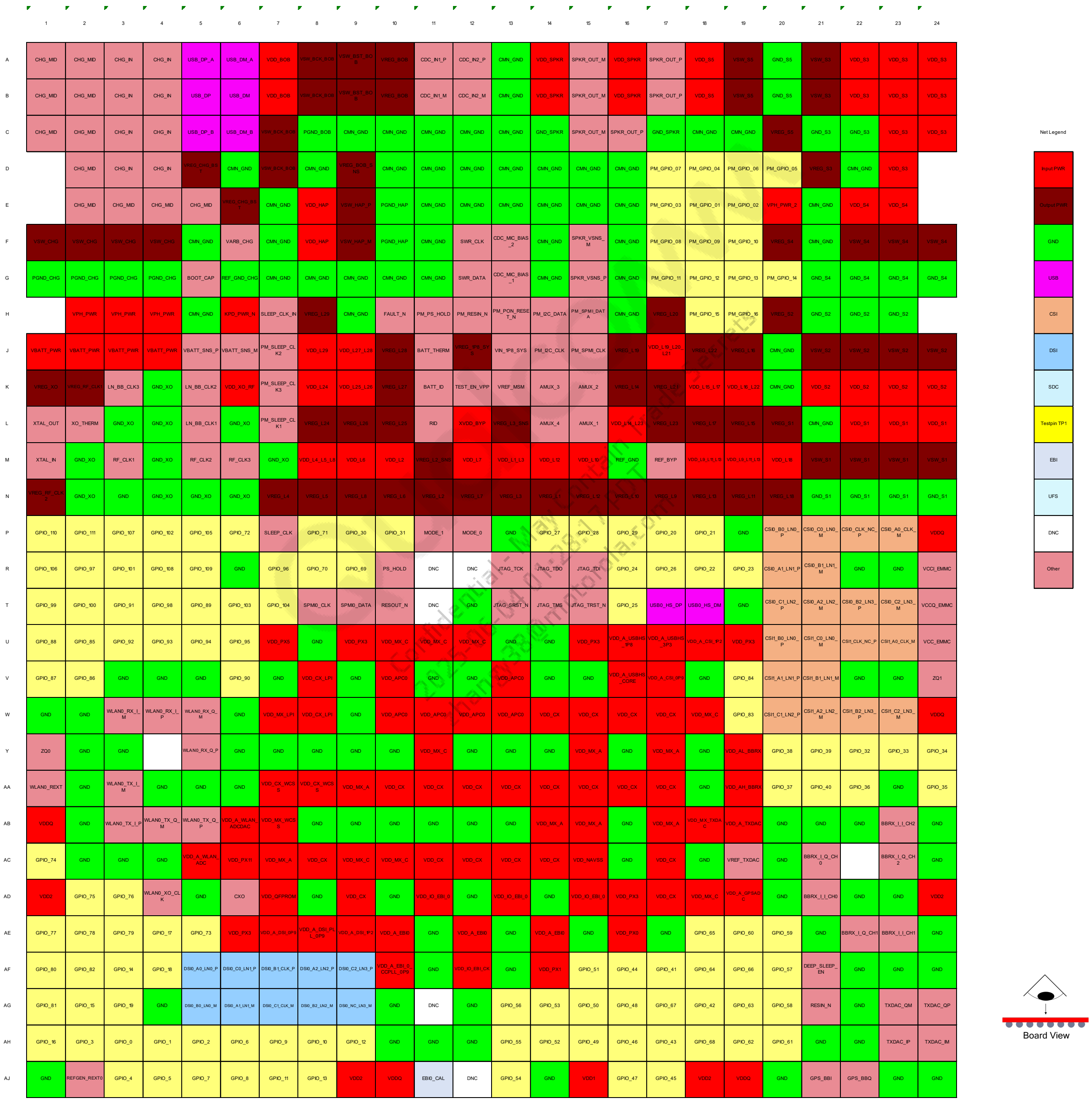

Figure 2-1 SW5100-0/SW5100P-0 bottom pin assignments

2.2.2 Pin descriptions – bottom (processor)

The pins are described in Table 2-2 through Table 2-4.

Table 2-2 SW5100-0/SW5100P-0 bottom pin descriptions – general pins

Pin #	Pin name	Pin characteristics		Functional description
		Voltage	Type	
AD21	BBRX_I_I_CH0	–	AI	Baseband receiver input, channel 0, in-phase
AE23	BBRX_I_I_CH1	–	AI	Baseband receiver input, channel 1, in-phase
AB23	BBRX_I_I_CH2	–	AI	Baseband receiver input, channel 2, in-phase
AC21	BBRX_I_Q_CH0	–	AI	Baseband receiver input, channel 0, quadrature
AE22	BBRX_I_Q_CH1	–	AI	Baseband receiver input, channel 1, quadrature
AC23	BBRX_I_Q_CH2	–	AI	Baseband receiver input, channel 2, quadrature
P23	CSI0_A0_CLK_M	CSI	AI, AO	MIPI CSI 0 (DPHY), differential clock – minus MIPI CSI 0 (CPHY), trio lane 0 – A
R20	CSI0_A1_LN1_P	CSI	AI, AO	MIPI CSI 0 (DPHY), differential lane 1 – plus MIPI CSI 0 (CPHY), trio lane 1 – A
T21	CSI0_A2_LN2_M	CSI	AI, AO	MIPI CSI 0 (DPHY), differential lane 2 – minus MIPI CSI 0 (CPHY), trio lane 2 – A
P20	CSI0_B0_LN0_P	CSI	AI, AO	MIPI CSI 0 (DPHY), differential lane 0 – plus MIPI CSI 0 (CPHY), trio lane 0 – B
R21	CSI0_B1_LN1_M	CSI	AI, AO	MIPI CSI 0 (DPHY), differential lane 1 – minus MIPI CSI 0 (CPHY), trio lane 1 – B
T22	CSI0_B2_LN3_P	CSI	AI, AO	MIPI CSI 0 (DPHY), differential lane 3 – plus MIPI CSI 0 (CPHY), trio lane 2 – B
P21	CSI0_C0_LN0_M	CSI	AI, AO	MIPI CSI 0 (DPHY), differential lane 0 – minus MIPI CSI 0 (CPHY), trio lane 0 – C
T20	CSI0_C1_LN2_P	CSI	AI, AO	MIPI CSI 0 (DPHY), differential lane 2 – plus MIPI CSI 0 (CPHY), trio lane 1 – C
T23	CSI0_C2_LN3_M	CSI	AI, AO	MIPI CSI 0 (DPHY), differential lane 3 – minus MIPI CSI 0 (CPHY), trio lane 2 – C
P22	CSI0_CLK_NC_P	CSI	AI, AO	MIPI CSI 0 (DPHY), differential clock – plus MIPI CSI 0 (CPHY), no connect
U23	CSI1_A0_CLK_M	CSI	AI, AO	MIPI CSI 1 (DPHY), differential clock – minus MIPI CSI 1 (CPHY), trio lane 0 – A
V20	CSI1_A1_LN1_P	CSI	AI, AO	MIPI CSI 1 (DPHY), differential lane 1 – plus MIPI CSI 1 (CPHY), trio lane 1 – A
W21	CSI1_A2_LN2_M	CSI	AI, AO	MIPI CSI 1 (DPHY), differential lane 2 – minus MIPI CSI 1 (CPHY), trio lane 2 – A
U20	CSI1_B0_LN0_P	CSI	AI, AO	MIPI CSI 1 (DPHY), differential lane 0 – plus MIPI CSI 1 (CPHY), trio lane 0 – B
V21	CSI1_B1_LN1_M	CSI	AI, AO	MIPI CSI 1 (DPHY), differential lane 1 – minus MIPI CSI 1 (CPHY), trio lane 1 – B
W22	CSI1_B2_LN3_P	CSI	AI, AO	MIPI CSI 1 (DPHY), differential lane 3 – plus MIPI CSI 1 (CPHY), trio lane 2 – B
U21	CSI1_C0_LN0_M	CSI	AI, AO	MIPI CSI 1 (DPHY), differential lane 0 – minus MIPI CSI 1 (CPHY), trio lane 0 – C

Table 2-2 SW5100-0/SW5100P-0 bottom pin descriptions – general pins (cont.)

Pin #	Pin name	Pin characteristics		Functional description
		Voltage	Type	
W20	CSI1_C1_LN2_P	CSI	AI, AO	MIPI CSI 1 (DPHY), differential lane 2 – plus MIPI CSI 1 (CPHY), trio lane 1 – C
W23	CSI1_C2_LN3_M	CSI	AI, AO	MIPI CSI 1 (DPHY), differential lane 3 – minus MIPI CSI 1 (CPHY), trio lane 2 – C
U22	CSI1_CLK_NC_P	CSI	AI, AO	MIPI CSI 1 (DPHY), differential clock plus MIPI CSI 1 (CPHY), no connect
AD6	CXO	PX_11	DI	Core crystal oscillator (digital 19.2 MHz system clock)
AF21	DEEP_SLEEP_EN	PX_0	DI	Enable pin for Deep Sleep
AF5	DSI0_A0_LN0_P	DSI	AI, AO	MIPI DSI 0 (DPHY), differential lane 0 – plus MIPI DSI 0 (CPHY), trio lane 0 – A
AG6	DSI0_A1_LN1_M	DSI	AI, AO	MIPI DSI 0 (DPHY), differential lane 1 – minus MIPI DSI 0 (CPHY), trio lane 1– A
AF8	DSI0_A2_LN2_P	DSI	AI, AO	MIPI DSI 0 (DPHY), differential lane 2 – plus MIPI DSI 0 (CPHY), trio lane 2– A
AG5	DSI0_B0_LN0_M	DSI	AI, AO	MIPI DSI 0 (DPHY), differential lane 0 – minus MIPI DSI 0 (CPHY), trio lane 0– B
AF7	DSI0_B1_CLK_P	DSI	AI, AO	MIPI DSI 0 (DPHY), differential clock – plus MIPI DSI 0 (CPHY), trio lane 1 – B
AG8	DSI0_B2_LN2_M	DSI	AI, AO	MIPI DSI 0 (DPHY), differential lane 2 – minus MIPI DSI 0 (CPHY), trio lane 2 – B
AF6	DSI0_C0_LN1_P	DSI	AI, AO	MIPI DSI 0 (DPHY), differential lane 1 – plus MIPI DSI 0 (CPHY), trio lane 0 – C
AG7	DSI0_C1_CLK_M	DSI	AI, AO	MIPI DSI 0 (DPHY), differential clock – minus MIPI DSI 0 (CPHY), trio lane 1 – C
AF9	DSI0_C2_LN3_P	DSI	AI, AO	MIPI DSI 0 (DPHY), differential lane 3 – plus MIPI DSI 0 (CPHY), trio lane 2 – C
AG9	DSI0_NC_LN3_M	DSI	AI, AO	MIPI DSI 0 (DPHY), differential lane 3 – minus MIPI DSI 0 (CPHY), no connect
AJ11	EBI0_CAL	–	AI	LPDDR4x calibration pin
AJ21	GPS_BBI	–	AI	GNSS receiver baseband input, in-phase
AJ22	GPS_BBQ	–	AI	GNSS receiver baseband input, quadrature phase
P12	MODE_0	P3	DI	Mode control bit 0
P11	MODE_1	P3	DI	Mode control bit 1
R10	PS_HOLD	P3	DO	Power – supply hold signal to PMIC
AJ2	REFGEN_REXT0	–	AI, AO	External calibration resistor for the reference PHY
AG21	RESIN_N	P0	DI	Reset input
T10	RESOUT_N	P3	DO	Reset output
P7	SLEEP_CLK	P3	DI	Sleep clock
T13	JTAG_SRST_N	P3	DI–PU	JTAG reset for debug
R13	JTAG_TCK	P3	DI–PU	JTAG clock input
R15	JTAG_TDI	P3	DIPU:nppdkp	JTAG data input
R14	JTAG_TDO	P3	DO–Z	JTAG data output
T14	JTAG_TMS	P3	DIPU:nppdkp	JTAG mode select input

Table 2-2 SW5100-0/SW5100P-0 bottom pin descriptions – general pins (cont.)

Pin #	Pin name	Pin characteristics		Functional description
		Voltage	Type	
T15	JTAG_TRST_N	P3	DIPD:nppukp	JTAG reset
AH24	TXDAC_IM	–	AI, AO	TXDAC in-phase – minus
AH23	TXDAC_IP	–	AI, AO	TXDAC in-phase – plus
AG23	TXDAC_QM	–	AI, AO	TXDAC quadrature – phase – minus
AG24	TXDAC_QP	–	AI, AO	TXDAC quadrature – phase – plus
T18	USB0_HS_DM	–	AI, AO	USB 0 high – speed data – minus
T17	USB0_HS_DP	–	AI, AO	USB 0 high – speed data – plus
AA1	WLAN0_REXT	–	AI, AO	WLAN chain external reference resistor pin
W3	WLAN0_RX_I_M	–	AI, AO	WLAN receive in-phase – minus
W4	WLAN0_RX_I_P	–	AI, AO	WLAN receive in-phase – plus
W5	WLAN0_RX_Q_M	–	AI, AO	WLAN receive quadrature – minus
Y5	WLAN0_RX_Q_P	–	AI, AO	WLAN receive quadrature – plus
AA3	WLAN0_TX_I_M	–	AI, AO	WLAN transmit – in-phase – minus
AB3	WLAN0_TX_I_P	–	AI, AO	WLAN transmit – in-phase – plus
AB4	WLAN0_TX_Q_M	–	AI, AO	WLAN transmit – quadrature – minus
AB5	WLAN0_TX_Q_P	–	AI, AO	WLAN transmit – quadrature – plus
AD4	WLAN0_XO_CLK	–	DI	WLAN reference clock
Y1	ZQ0	–	AI	Pin for ZQ0 calibration
V24	ZQ1	–	AI	Pin for ZQ1 calibration
T8	SPMI0_CLK	–	B	Slave and PBUS interface for PMICs – clock
T9	SPMI0_DATA	–	B	Slave and PBUS interface for PMICs – data
AC19	VREF_TXDAC	–	AI, AO	Transmitter DAC voltage reference

NOTE GPIO pins can support multiple functions. To assign GPIOs to particular functions (such as the options listed in the preceding table), designers must identify all their application's requirements and map each GPIO to its function – carefully avoiding conflicts in GPIO assignments. See the following table for a list of all supported functions for each GPIO.

Device designers must examine each GPIO's external connection and programmed configuration, and take steps necessary to avoid excessive leakage current. Combinations of the following factors must be controlled properly:

- GPIO configuration
  - Input vs. output
  - Pull-up or pull-down
- External connections
  - Unused inputs
  - Connections to high-impedance (tri-state) outputs
  - Connections to external devices that may not be attached

To help designers define their products' GPIO assignments, QTI provides an Excel spreadsheet that includes a worksheet for all SW5100-0/SW5100P-0 GPIOs (in numeric order), pin numbers, pin voltages, pull states, and available configurations.

NOTE Click the following link to download the pin assignment and GPIO configuration spreadsheet from the CreatePoint website.

https://createpoint.qti.qualcomm.com/search/contentdocument/stream/dcn/80-24633-1A

After successfully logging in, the document is downloaded. Make this document a favorite to be notified of any changes.

Table 2-3 SW5100-0/SW5100P-0 bottom pin descriptions – general-purpose input/output ports

Pin #	Pin name	Wakeup function		Configurable functions	Pin characteristics		Functional description
		LPI wakeup	MPM wakeup		Voltage	Type	
AH3	GPIO_0	–	Y	QUP0_L0[2]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE2 lane 0: UART_CTS/SPI_MISO/ I2C_SDA
AH4	GPIO_1	–	–	QUP0_L1[2]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE2 lane 1: UART_RFR/SPI_MOSI/ I2C_SCL
AH5	GPIO_2	–	–	QUP0_L2[2]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE2 lane 2: UART_TX/SPI_CLK
AH2	GPIO_3	–	Y	QUP0_L3[2]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE2 lane 3: UART_RX/SPI_CS0
AJ3	GPIO_4	–	Y	QUP0_L0[0]	PX3/BPP	B-PD:nppukp	Configurable I/O QUP 0 SE0 lane 0: UART_CTS/SPI_MISO/ I2C_SDA
AJ4	GPIO_5	–	–	QUP0_L1[0]	PX3/BPP	B-PD:nppukp	Configurable I/O QUP 0 SE0 lane 1: UART_RFR/SPI_MOSI/ I2C_SCL
AH6	GPIO_6	–	Y	QUP0_L2[0]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE0 lane 2: UART_TX/SPI_CLK
AJ5	GPIO_7	Y	Y	QUP0_L3[0]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE0 lane 3: UART_RX/SPI_CS0
AJ6	GPIO_8	–	Y	QUP0_L4[2]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE2 lane 4: SPI_CS1
AH7	GPIO_9	–	Y	QUP0_L5[2]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE2 lane 5: SPI_CS2
AH8	GPIO_10	–	Y	QUP0_L0[1]	PX3/BPP	B-PD:nppukp	Configurable I/O QUP 0 SE1 lane 0: UART_CTS/SPI_MISO/ I2C_SDA

Table 2-3 SW5100-0/SW5100P-0 bottom pin descriptions – general-purpose input/output ports (cont.)

Pin #	Pin name	Wakeup function		Configurable functions	Pin characteristics		Functional description
		LPI wakeup	MPM wakeup		Voltage	Type	
AJ7	GPIO_11	–	–	QUP0_L1[1]	PX3/BPP	B-PD:nppukp	Configurable I/O QUP 0 SE1 lane 1: UART_RFR/SPI_MOSI/ I2C_SCL
AH9	GPIO_12	–	–	QUP0_L2[1]	PX3/BPP	B-PD:nppukp	Configurable I/O QUP 0 SE1 lane 2: UART_TX/SPI_CLK
AJ8	GPIO_13	Y	Y	QUP0_L3[1]	PX3/BPP	B-PD:nppukp	Configurable I/O QUP 0 SE1 lane 3: UART_RX/SPI_CS0
AF3	GPIO_14	–	Y	QUP0_L0[3]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE3 lane 0: UART_CTS/SPI_MISO/ I2C_SDA
AG2	GPIO_15	–	–	QUP0_L1[3]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE3 lane 1: UART_RFR/SPI_MOSI/ I2C_SCL
AH1	GPIO_16	–	Y	QUP0_L2[3]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE3 lane 2: UART_TX/SPI_CLK
AE4	GPIO_17	–	Y	QUP0_L3[3]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE3 lane 3: UART_RX/SPI_CS0
AF4	GPIO_18	–	Y	QUP0_L4[3]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE3 lane 4: SPI_CS1
AG3	GPIO_19	Y	Y	QUP0_L5[3]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE3 lane 5: SPI_CS2
P17	GPIO_20	Y	Y	QUP0_L0[4]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE4 lane 0: UART_CTS/SPI_MISO/ I2C_SDA
P18	GPIO_21	Y	–	QUP0_L1[4]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE4 lane 1: UART_RFR/ SPI_MOSI_I2C_SCL

Table 2-3 SW5100-0/SW5100P-0 bottom pin descriptions – general-purpose input/output ports (cont.)

Pin #	Pin name	Wakeup function		Configurable functions	Pin characteristics		Functional description
		LPI wakeup	MPM wakeup		Voltage	Type	
R18	GPIO_22	Y	–	QUP0_L2[4]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE4 lane 2: UART_TX/SPI_CLK
R19	GPIO_23	Y	Y	QUP0_L3[4]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE4 lane 3: UART_RX/SPI_CS0
R16	GPIO_24	Y	Y	QUP0_L0[6]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE6 lane 0: UART_CTS/SPI_MISO/ I2C_SDA/I3C_SDA
T16	GPIO_25	Y	Y	QUP0_L1[6]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE6 lane 1: UART_RFR/SPI_MOSI/ I2C_SCL/I3C_SCL
R17	GPIO_26	–	Y	QUP0_L0[5]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE5 lane 0: UART_CTS/SPI_MISO/ I2C_SDA/I3C_SDA
P14	GPIO_27	–	–	QUP0_L1[5]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE5 lane 1: UART_RFR/SPI_MOSI/ I2C_SCL/I3C_SCL
P15	GPIO_28	–	–	QUP0_L2[5]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE5 lane 2: UART_TX/SPI_CLK
P16	GPIO_29	–	Y	QUP0_L3[5]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE5 lane 3: UART_RX/SPI_CS0
P9	GPIO_30	–	–	QUP0_L2[6]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE6 lane 2: UART_TX/SPI_CLK
P10	GPIO_31	–	Y	QUP0_L3[6]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE6 lane 3: UART_RX/SPI_CS0
Y22	GPIO_32	–	–	CAM_MCLK0	PX3	B-PD:nppukp DO	Configurable I/O Camera master clock 0
Y23	GPIO_33	–	Y	CAM_MCLK1	PX3	B-PD:nppukp DO	Configurable I/O Camera master clock 1

Table 2-3 SW5100-0/SW5100P-0 bottom pin descriptions – general-purpose input/output ports (cont.)

Pin #	Pin name	Wakeup function		Configurable functions	Pin characteristics		Functional description
		LPI wakeup	MPM wakeup		Voltage	Type	
Y24	GPIO_34	–	Y	 CAM_MCLK2	PX3	B-PD:nppukp DO	Configurable I/O Camera master clock 2
AA24	GPIO_35	–	–	 QUP0_L4[5]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE5 lane 4: SPI_CS1
AA22	GPIO_36	–	Y	 QUP0_L5[5]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE5 lane 5: SPI_CS2
AA20	GPIO_37	–	–	 CCI_I2C_SDA0	PX3	B-PD:nppukp B	Configurable I/O Dedicated camera control interface I2C0 data
Y20	GPIO_38	–	–	 CCI_I2C_SCL0	PX3	B-PD:nppukp DO	Configurable I/O Dedicated camera control interface I2C0 clock
Y21	GPIO_39	–	Y	 CCI_I2C_SDA1	PX3	B-PD:nppukp B	Configurable I/O Dedicated camera control interface I2C1 data
AA21	GPIO_40	–	Y	 CCI_I2C_SCL1	PX3	B-PD:nppukp DO	Configurable I/O Dedicated camera control interface I2C1 clock
AF17	GPIO_41	–	Y	 GRFC14	PX3	B-PD:nppukp DO	Configurable I/O Generic RF controller bit 14
AG18	GPIO_42	–	Y	 GRFC15	PX3	B-PD:nppukp DO	Configurable I/O Generic RF controller bit 15
AH17	GPIO_43	–	–	 GRFC0 BOOT_CONFIG[0]	PX3	B-PD:nppukp DO	Configurable I/O Generic RF controller bit 0
AF16	GPIO_44	–	–	 GRFC1 BOOT_CONFIG[1]	PX3	B-PD:nppukp DO	Configurable I/O Generic RF controller bit 1
AJ17	GPIO_45	–	Y	 GRFC2 BOOT_CONFIG[2]	PX3	B-PD:nppukp DO	Configurable I/O Generic RF controller bit 2

Table 2-3 SW5100-0/SW5100P-0 bottom pin descriptions – general-purpose input/output ports (cont.)

Pin #	Pin name	Wakeup function		Configurable functions	Pin characteristics		Functional description
		LPI wakeup	MPM wakeup		Voltage	Type	
AH16	GPIO_46	–	–	 GRFC3 BOOT_CONFIG[3]	PX3	B-PD:nppukp DO	Configurable I/O Generic RF controller bit 3
AJ16	GPIO_47	–	Y	 NAV_GPIO0	PX3	B-PD:nppukp B	Configurable I/O Generic IO for GNSS
AG16	GPIO_48	–	Y	 NAV_GPIO1	PX3	B-PD:nppukp B	Configurable I/O Generic IO for GNSS
AH15	GPIO_49	–	Y	 NAV_GPIO2	PX3	B-PD:nppukp B	Configurable I/O Generic IO for GNSS
AG15	GPIO_50	–	–	 GRFC7 BOOT_CONFIG[4]	PX3	B-PD:nppukp DO	Configurable I/O Generic RF controller bit 7
AF15	GPIO_51	–	–	 GRFC8 BOOT_CONFIG[5]	PX3	B-PD:nppukp DO	Configurable I/O Generic RF controller bit 8
AH14	GPIO_52	–	Y	 GRFC9 BOOT_CONFIG[6]	PX3	B-PD:nppukp DO	Configurable I/O Generic RF controller bit 9
AG14	GPIO_53	–	–	 GRFC10 BOOT_CONFIG[7]	PX3	B-PD:nppukp DO	Configurable I/O Generic RF controller bit 10
AJ13	GPIO_54	–	–	 GRFC11 BOOT_CONFIG[8]	PX3	B-PD:nppukp DO	Configurable I/O Generic RF controller bit 11
AH13	GPIO_55	–	Y	 GRFC12	PX3	B-PD:nppukp DO	Configurable I/O Generic RF controller bit 12
AG13	GPIO_56	–	Y	 GRFC13	PX3	B-PD:nppukp DO	Configurable I/O Generic RF controller bit 13

Table 2-3 SW5100-0/SW5100P-0 bottom pin descriptions – general-purpose input/output ports (cont.)

Pin #	Pin name	Wakeup function		Configurable functions	Pin characteristics		Functional description
		LPI wakeup	MPM wakeup		Voltage	Type	
AF20	GPIO_57	–	–	 RFFE1_DATA BOOT_CONFIG[9]	PX3	B-PD:nppukp B	Configurable I/O RF front-end 1 interface data
AG20	GPIO_58	–	–	 RFFE1_CLK	PX3	B-PD:nppukp DO	Configurable I/O RF front-end 1 interface clock
AE20	GPIO_59	–	–	 RFFE2_DATA BOOT_CONFIG[10]	PX3	B-PD:nppukp B	Configurable I/O RF front-end 2 interface data
AE19	GPIO_60	–	–	 RFFE2_CLK	PX3	B-PD:nppukp DO	Configurable I/O RF front-end 2 interface clock
AH20	GPIO_61	–	Y	 RFFE3_DATA BOOT_CONFIG[11]	PX3	B-PD:nppukp B	Configurable I/O RF front-end 3 interface data
AH19	GPIO_62	–	Y	 RFFE3_CLK	PX3	B-PD:nppukp DO	Configurable I/O RF front-end 3 interface clock
AG19	GPIO_63	–	Y	 RFFE4_DATA	PX3	B-PD:nppukp B	Configurable I/O RF front-end 4 interface data
AF18	GPIO_64	–	Y	 RFFE4_CLK	PX3	B-PD:nppukp DO	Configurable I/O RF front-end 4 interface clock
AE18	GPIO_65	–	–	 RFFE5_DATA	PX3	B-PD:nppukp B	Configurable I/O RF front-end 5 interface data
AF19	GPIO_66	–	Y	 RFFE5_CLK	PX3	B-PD:nppukp DO	Configurable I/O RF front-end 5 interface clock
AG17	GPIO_67	–	Y		PX3	B-PD:nppukp	Configurable I/O
AH18	GPIO_68	–	Y	 GSM0_TX_PHASE_D	PX3	B-PD:nppukp	Configurable I/O GSM 0 transmit phase adjust data bit

Table 2-3 SW5100-0/SW5100P-0 bottom pin descriptions – general-purpose input/output ports (cont.)

Pin #	Pin name	Wakeup function		Configurable functions	Pin characteristics		Functional description
		LPI wakeup	MPM wakeup		Voltage	Type	
R9	GPIO_69	–	–	 UIM0_DATA	PX5	B-PD:nppukp B	Configurable I/O UIM0 data (1.8 V)
R8	GPIO_70	–	–	 UIM0_CLK	PX5	B-PD:nppukp DO	Configurable I/O UIM0 clock (1.8 V)
P8	GPIO_71	–	–	 UIM0_RESET	PX5	B-PD:nppukp DO	Configurable I/O UIM0 reset (1.8 V)
P6	GPIO_72	–	Y	 UIM0_PRESENT	PX5	B-PD:nppukp DI	Configurable I/O UIM0 presence detection
AE5	GPIO_73	–	Y	 MDP_VSYNC_P	PX3/BPP	B-PD:nppukp DI	Configurable I/O MDP vertical sync – primary
AC1	GPIO_74	–	Y	 SDC3_DATA_3	PX3	B-PD:nppukp B	Configurable I/O Secure digital controller data bit 3
AD2	GPIO_75	–	–	 SDC3_DATA_2	PX3	B-PD:nppukp B	Configurable I/O Secure digital controller data bit 2
AD3	GPIO_76	–	Y	 SDC3_DATA_1	PX3	B-PD:nppukp B	Configurable I/O Secure digital controller data bit 1
AE1	GPIO_77	–	–	 SDC3_DATA_0	PX3	B-PD:nppukp B	Configurable I/O Secure digital controller data bit 0
AE2	GPIO_78	–	Y	 SDC3_CMD	PX3	B-PD:nppukp B	Configurable I/O Secure digital controller command
AE3	GPIO_79	–	Y	 SDC3_CLK	PX3	B-PD:nppukp DO	Configurable I/O Secure digital controller clock
AF1	GPIO_80	Y	Y	 QUP0_L6[3]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE3 lane 6: SPI_CS3
AG1	GPIO_81	–	–	 WCI_UART_TXD	PX3	B-PD:nppukp DO	Configurable I/O Wireless co-existence interface UART TX

Table 2-3 SW5100-0/SW5100P-0 bottom pin descriptions – general-purpose input/output ports (cont.)

Pin #	Pin name	Wakeup function		Configurable functions	Pin characteristics		Functional description
		LPI wakeup	MPM wakeup		Voltage	Type	
AF2	GPIO_82	–	Y	 WCI_UART_RXD	PX3	B-PD:nppukp DI	Configurable I/O Wireless co-existence interface UART RX
W19	GPIO_83	–	Y	 FORCED_USB_ BOOT_POL_SEL	PX3	B-PD:nppukp DI	Configurable I/O Polarity select for forced USB Boot
V19	GPIO_84	–	Y	 FORCED_USB_BOOT	PX3	B-PD:nppukp DI	Configurable I/O Forced USB boot for emergency download
U2	GPIO_85	–	–	 LPI_GPIO_0	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 0
V2	GPIO_86	Y	Y	 LPI_GPIO_1	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 1
V1	GPIO_87	–	Y	 LPI_GPIO_2	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 2
U1	GPIO_88	–	–	 LPI_GPIO_3	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 3
T5	GPIO_89	Y	–	 LPI_GPIO_4	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 4
V6	GPIO_90	–	Y	 LPI_GPIO_5	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 5
T3	GPIO_91	–	–	 LPI_GPIO_6	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 6
U3	GPIO_92	–	–	 LPI_GPIO_7	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 7
U4	GPIO_93	–	–	 LPI_GPIO_8	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 8
U5	GPIO_94	–	–	 LPI_GPIO_9	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 9

Table 2-3 SW5100-0/SW5100P-0 bottom pin descriptions – general-purpose input/output ports (cont.)

Pin #	Pin name	Wakeup function		Configurable functions	Pin characteristics		Functional description
		LPI wakeup	MPM wakeup		Voltage	Type	
U6	GPIO_95	Y	Y	LPI_GPIO_10	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 10
R7	GPIO_96	Y	Y	LPI_GPIO_11	PX3/BPP	B-PD:nppukp	Configurable I/O Low power island I/O 11
R2	GPIO_97	–	Y	LPI_GPIO_12	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 12
T4	GPIO_98	Y	Y	LPI_GPIO_13	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 13
T1	GPIO_99	–	–	LPI_GPIO_14	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 14
T2	GPIO_100	Y	–	LPI_GPIO_15	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 15
R3	GPIO_101	–	Y	QUP0_L0_7_MIRA LPI_GPIO_16	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE7 lane 0: UART_CTS/SPI_MISO/ I2C_SDA Low power island I/O 16
P4	GPIO_102	–	Y	QUP0_L1_7_MIRA LPI_GPIO_17	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE7 lane 1: UART_RFR/SPI_MOSI/ I2C_SCL Low power island I/O 17
T6	GPIO_103	–	–	LPI_GPIO_18	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 18
T7	GPIO_104	Y	Y	QUP0_L2_7_MIRA LPI_GPIO_19 LPI_QUP_L0[0]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE7 lane 2: UART_TX/SPI_CLK Low power island I/O 19 LPI QUP SE0 lane 0: I3C_SDA

Table 2-3 SW5100-0/SW5100P-0 bottom pin descriptions – general-purpose input/output ports (cont.)

Pin #	Pin name	Wakeup function		Configurable functions	Pin characteristics		Functional description
		LPI wakeup	MPM wakeup		Voltage	Type	
P5	GPIO_105	–	Y	 QUP0_L3_7_MIRA LPI_GPIO_20 LPI_QUP_L1[0]	PX3	B-PD:nppukp	Configurable I/O QUP 0 SE7 lane 3: UART_RX/SPI_CS0 Low power island I/O 20 LPI QUP SE0 lane 1: I3C_SCL
R1	GPIO_106	Y	–	 LPI_GPIO_21 LPI_QUP_L0[1]	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 21 LPI QUP SE1 lane 0: SPI_MISO/I2C_SDA/ I3C_SDA
P3	GPIO_107	–	–	 LPI_GPIO_22 LPI_QUP_L1[1]	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 22 LPI QUP SE1 lane 1: SPI_MOSI/I2C_SCL/ I3C_SCL
R4	GPIO_108	–	–	 LPI_GPIO_23 LPI_QUP_L2[5] LPI_QUP_L0[5]	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 23 LPI QUP SE5 lane 2: UART_TX LPI QUP SE5 lane 0: I2C_SDA
R5	GPIO_109	Y	Y	 LPI_GPIO_24 LPI_QUP_L3[5] LPI_QUP_L1[5]	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 24 LPI QUP SE5 lane 3: UART_RX LPI QUP SE5 lane 1: I2C_SCL
P1	GPIO_110	–	–	 LPI_GPIO_25 LPI_QUP_L2[1] LPI_QUP_L2[6]	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 25 LPI QUP SE1 lane 2: SPI_CLK LPI QUP SE6 lane 2: UART_TX
P2	GPIO_111	Y	Y	 LPI_GPIO_26 LPI_QUP_L3[1] LPI_QUP_L3[6]	PX3	B-PD:nppukp	Configurable I/O Low power island I/O 26 LPI QUP SE1 lane 3: SPI_CS LPI QUP SE6 lane 3: UART_RX

Table 2-4 SW5100-0/SW5100P-0 bottom pin descriptions – NC, ground, and power-supply pins

Pin #	Pin name	Functional description
AJ12, T11, AG11, R11, R12	NC	No connection
R24	VCCI_EMMC	Power for eMMC – core IO
T24	VCCQ_EMMC	Power for memory I/O
U24	VCC_EMMC	Power for eMMC – control
AJ15	VDD1	Power for DDR memory core – 1.8 V
AD1, AD24, AJ18, AJ9	VDD2	Power for DDR memory core – 1.2 V
AA19	VDD_AH_BBRX	Power for analog circuits: high voltage
Y19	VDD_AL_BBRX	Power for analog circuits: low voltage
V17	VDD_A_CSI_0P9	Power for the CSI – 0.9 V circuits
U18	VDD_A_CSI_1P2	Power for the CSI – 1.2 V circuits
AE7	VDD_A_DSI_0P9	Power for the DSI – 0.9 V circuits
AE9	VDD_A_DSI_1P2	Power for the DSI – 1.2 V circuits
AE8	VDD_A_DSI_PLL_0P9	Power for the DSI PLL circuits
AE10, AE12, AE14	VDD_A_EBI0	Power for EBI circuits
AF10	VDD_A_EBI_0_ CCPLL_0P9	Power for EBI PLL circuits
AD19	VDD_A_GPSADC	Power for the GPS ADC circuits
AB19	VDD_A_TXDAC	Power for TXDAC modem circuits
AB6	VDD_A_WLAN_ ADCDAC	Analog supply for WLAN DAC
AC5	VDD_A_WLAN_ADC	Analog supply for WLAN ADC
AA10, AA11, AA12, AA13, AA14, AA15, AA16, AA17, AC11, AC12, AC13, AC14, AC17, AC8, AD17, AD9, W14, W15, W16, W17	VDD_CX	Power for SoC core logic domain
AD11, AD13, AD15	VDD_IO_EBI_0	Power for the EBI I/O circuits
AF12	VDD_IO_EBI_CK	Power for the EBI I/O clock circuits
AA9, AB14, AB15, AB17, AC7, Y15, Y17	VDD_MX_A	Power rail for always ON memory
AC10, AC9, AD18, U10, U11, U12, W18, Y11	VDD_MX_C	Collapsible memory rail for GPU/Camera/VIDEO/APC/ Modem
AE16	VDD_PX0	Special pins for the EBI I/O circuits Power and DEEP_SLEEP feature
AF14	VDD_PX1	Power for the EBI I/O circuits
AD16, AE6, U15, U19, U9	VDD_PX3	Power for the pin group 3 – most I/O pins
U7	VDD_PX5	Power for the pin group 5 – UIM0 pins
AC6	VDD_PX11	Power for the CXO pins
AB1, AJ10, AJ19, P24, W24	VDDQ	Power for memory I/O
V10, V13, W10, W11, W12, W13	VDD_APC0	Power for the application processor
V8, W8	VDD_CX_LPI	Sensor and LPASS island logic domain
W7	VDD_MX_LPI	Sensor and LPASS island logic domain
AB18	VDD_MX_TXDAC	Power for TXDAC memory circuits

Table 2-4 SW5100-0/SW5100P-0 bottom pin descriptions – NC, ground, and power-supply pins (cont.)

Pin #	Pin name	Functional description
AC15	VDD_NAVSS	Power for navigation sub-system
AD7	VDD_QFPROM	Power for programming the QFPROM
V16	VDD_A_USBHS_ CORE	Power for the USB HS 0.9 V circuits – USB core
U16	VDD_A_USBHS_1P8	Power for the USB HS 1.8 V circuits – low voltage PHY
U17	VDD_A_USBHS_3P3	Power for the USB HS3.3 V circuits – high voltage PHY
AA7, AA8	VDD_CX_WCSS	Digital rail for WLAN island
AB7	VDD_MX_WCSS	Memory rail for WLAN merged with WCSS_CX rail on board
AA18, AA2, AA23, AA4, AA5, AA6, AB10, AB11, AB12, AB13, AB16, AB2, AB20, AB21, AB22, AB24, AB8, AB9, AC16, AC18, AC2, AC20, AC24, AC3, AC4, AD10, AD12, AD14, AD20, AD22, AD23, AD5, AD8, AE11, AE13, AE15, AE17, AE21, AE24, AF11, AF13, AF22, AF23, AF24, AG10, AG12, AG22, AG4, AH10, AH11, AH12, AH21, AH22, AJ1, AJ14, AJ20, AJ23, AJ24, N3, P13, P19, R22, R23, R6, T12, T19, U13, U14, U8, V11, V12, V14, V15, V18, V22, V23, V3, V4, V5, V7, V9, W1, W2, W6, W9, Y10, Y12, Y13, Y14, Y16, Y18, Y2, Y3, Y6, Y7, Y8, Y9	GND	Ground

2.2.3 Pin descriptions – bottom (PMIC)

Descriptions of all pins are presented in the following tables, organized by functional group:

Table 2-5: Input power management

Table 2-6: Output power management

Table 2-7: General housekeeping

Table 2-8: User interface

Table 2-9: IC-level interfaces

Table 2-10: General-purpose input/output

Table 2-11: Audio

Table 2-12: Common ground pins

Table 2-5 Pin descriptions – input power management functions

Pin #	Pin name and/or function	Pin type [a]	Functional description
Charger interface			
A3, A4, B3, B4, C3, C4, D3, D4	CHG_IN	PI	Charger input power; USB VBUS input or wireless charger input
B5	USB_DP	AI, AO	USB data plus for power source detection only; QTI SoC handles data transactions.

Table 2-5 Pin descriptions – input power management functions (cont.)

Pin #	Pin name and/or function	Pin type [a]	Functional description
B6	USB_DM	AI, AO	USB data minus for power source detection only; QTI SoC handles data transactions.
A5 [b]	USB_DP_A	AI, AO	UMUX channel A (USB_DP). Connect to USB0_HS_DP of SW5100.
A6 [b]	USB_DM_A	AI, AO	UMUX channel A (USB_DM). Connect to USB0_HS_DM of SW5100
C5 [b]	USB_DP_B	AI, AO	UMUX channel B (DEBUG_UART_TX). Connect to GPIO_30 of SW5100
C6 [b]	USB_DM_B	AI, AO	UMUX channel B (DEBUG_UART_RX). Connect to GPIO_31 of SW5100
Switching charger (SCHG)			
H2, H3, H4	VPH_PWR	PI, PO	Primary system supply node, switching charger (SCHG) regulated node.
E20	VPH_PWR_2	PI	PMIC input power node generated by either the charger or battery.
G5	BOOT_CAP	AO	HS FET driver bootstrap capacitor connection
F1, F2, F3, F4	VSW_CHG	PO	Charger SMPS switching node.
G1, G2, G3, G4	PGND_CHG	GNDP	Specific ground for the SCHG. layout considerations must be made for this pin.
A1, A2, B1, B2, C1, C2, D2, E2, E3, E4, E5	CHG_MID	AO	Mid joint point of FP FET and charger buck input. Layout consideration must be made for this pin.
G6	REF_GND_CHG	GNDP	Dedicated ground for the charger-specific master bandgap. Special considerations must be made to ensure that this ground is properly connected on the PCB.
D5, E6	VREG_CHG_BST	PI, PO	Charger boost output
Qualcomm battery gauge/battery interface			
K11	BATT_ID	AI	Battery ID input to the ADC. It can be used for missing battery detection.
J11	BATT_THERM	AI	Battery temperature input to ADC for measuring the pack temperature. It is used for charger safe operation and Qualcomm battery gauge.
F6	VARB_CHG	AO	Power source arbitration circuit output and must not be used for other purposes.
J1, J2, J3, J4	VBATT_PWR	PI, PO	Battery voltage node, connects to BATFET. Output is for charging, and input is for all other operations.
J5	VBATT_SNS_P	AI	Battery voltage sense input plus. Connect to the battery positive remote sense node or connect this directly to the battery positive node.
J6	VBATT_SNS_M	AI	Battery voltage sense input minus. Connect to the battery negative remote sense node or connect this directly to the battery negative node.

[a] See Table 2-1 for pin voltage and type definitions.
[b] See the QTI reference schematics for connection details.

Table 2-6 Pin descriptions – output power management functions

Pin #	Pin name and/or function	Pin type [a]	Functional description
High-frequency buck SMPS circuits			
L22, L23, L24	VDD_S1	PI	S1 SMPS supply power input
N21, N22, N23, N24	GND_S1	GNDP	S1 SMPS power ground
M21, M22, M23, M24	VSW_S1	PO	S1 SMPS switch node
L20	VREG_S1	AI	S1 SMPS sense input
K21, K22, K23, K24	VDD_S2	PI	S2 SMPS supply power input
H21, H22, H23	GND_S2	GNDP	S2 SMPS power ground
J21, J22, J23, J24	VSW_S2	PO	S2 SMPS switch node
H20	VREG_S2	AI	S2 SMPS sense input
A22, A23, A24, B22, B23, B24, C23, C24, D23	VDD_S3	PI	S3 SMPS supply power input
C21, C22	GND_S3	GNDP	S3 SMPS power ground
A21, B21	VSW_S3	PO	S3 SMPS switch node
D21	VREG_S3	AI	S3 SMPS sense input
E22, E23	VDD_S4	PI	S4 SMPS supply power input
G21, G22, G23, G24	GND_S4	GNDP	S4 SMPS power ground
F22, F23, F24	VSW_S4	PO	S4 SMPS switch node
F20	VREG_S4	AI	S4 SMPS sense input
A18, B18	VDD_S5	PI	S5 SMPS supply power input
A20, B20	GND_S5	GNDP	S5 SMPS power ground
A19, B19	VSW_S5	PO	S5 SMPS switch node
C20	VREG_S5	AI	S5 SMPS sense input
Low dropout regulator (LDO) circuits			
N14	VREG_L1	PO	L1 LDO regulated output
N11	VREG_L2	PO	L2 LDO regulated output
M11	VREG_L2_SNS	AI	L2 LDO SNS input
N13	VREG_L3	PO	L3 LDO regulated output
L13	VREG_L3_SNS	AI	L3 LDO SNS input
N7	VREG_L4	PO	L4 LDO regulated output
N8	VREG_L5	PO	L5 LDO regulated output
N10	VREG_L6	PO	L6 LDO regulated output
N12	VREG_L7	PO	L7 LDO regulated output
N9	VREG_L8	PO	L8 LDO regulated output
N17	VREG_L9	PO	L9 LDO regulated output
N16	VREG_L10	PO	L10 LDO regulated output
N19	VREG_L11	PO	L11 LDO regulated output
N15	VREG_L12	PO	L12 LDO regulated output
N18	VREG_L13	PO	L13 LDO regulated output
K16	VREG_L14	PO	L14 LDO regulated output

Table 2-6 Pin descriptions – output power management functions (cont.)

Pin #	Pin name and/or function	Pin type [a]	Functional description
L19	VREG_L15	PO	L15 LDO regulated output
J19	VREG_L16	PO	L16 LDO regulated output
L18	VREG_L17	PO	L17 LDO regulated output
N20	VREG_L18	PO	L18 LDO regulated output
J16	VREG_L19	PO	L19 LDO regulated output
H17	VREG_L20	PO	L20 LDO regulated output
K17	VREG_L21	PO	L21 LDO regulated output
J18	VREG_L22	PO	L22 LDO regulated output
L17	VREG_L23	PO	L23 LDO regulated output
L8	VREG_L24	PO	L24 LDO regulated output
L10	VREG_L25	PO	L25 LDO regulated output
L9	VREG_L26	PO	L26 LDO regulated output
K10	VREG_L27	PO	L27 LDO regulated output
J10	VREG_L28	PO	L28 LDO regulated output
H8	VREG_L29	PO	L29 LDO regulated output
M13	VDD_L1_L3	PI	Power supply for L1 and L3 LDOs
M10	VDD_L2	PI	Power supply for L2 LDO
M8	VDD_L4_L5_L8	PI	Power supply for L4, L5, and L8 LDOs
M9	VDD_L6	PI	Power supply for L6 LDO
M12	VDD_L7	PI	Power supply for L7 LDO
M18, M19	VDD_L9_L11_L13	PI	Power supply for L9, L11, and L13 LDOs
M15	VDD_L10	PI	Power supply for L10 LDOs
M14	VDD_L12	PI	Power supply for L12 LDOs
L16	VDD_L14_L23	PI	Power supply for L14 and L23 LDOs
K18	VDD_L15_L17	PI	Power supply for L15 and L17 LDOs
K19	VDD_L16_L22	PI	Power supply for L16 and L22 LDOs
M20	VDD_L18	PI	Power supply for L18 LDOs
J17	VDD_L19_L20_L21	PI	Power supply for L19, L20, and L21 LDOs
K8	VDD_L24	PI	Power supply for L24 LDOs
K9	VDD_L25_L26	PI	Power supply for L25 and L26 LDOs
J9	VDD_L27_L28	PI	Power supply for L27 and L28 LDOs
J8	VDD_L29	PI	Power supply for L29 LDOs
Buck-or-boost (BoB) circuits			
A8, B8, C7, D7	VSW_BCK_BOB	PO	Buck SMPS switch node
A9, B9	VSW_BST_BOB	PO	Boost SMPS switch node
A10, B10	VREG_BOB	PO	Power supply from BoB
D9	VREG_BOB_SNS	AI	Regulated BoB output sense line
A7, B7	VDD_BOB	PI	BoB input power; connects to VPH_PWR
C8	PGND_BOB	GNDP	BoB power ground

[a] See Table 2-1 for pin voltage and type definitions.

Table 2-7 Pin descriptions – general housekeeping functions

Pin #	Pin name and/or function	Pin characteristics [a]		Functional description
		Voltage	Type	
HK/XO ADC and analog multiplexer circuits				
L2	XO_THERM	–	AI	XO thermistor resistor input
L15	AMUX_1	–	AI	CAM_THERM: Place the thermistor near camera sensors/connector. For camera less design, place near wireless charger.
K15	AMUX_2	–	AI	PA_THERM: Place the thermistor near RF power amplifier
K14	AMUX_3	–	AI	SKIN_THERM: Place near back cover or on HRM sensor FPC
L14	AMUX_4	–	AI	OPTION_PIN, pull down with resistor value used for setting power-on options. This pin cannot be used for any other purpose. [b]
Sleep clock circuits				
H7	SLEEP_CLK_IN	VREG_1P8_SYS	DI	External 32.768 kHz TXCO (temperature compensated clock oscillator) input for accurate RTC clock.
L7	PM_SLEEP_CLK1	VREG_1P8_SYS	DO	Sleep clock output to SW5100
J7	PM_SLEEP_CLK2	VREG_1P8_SYS	DO	Sleep clock output to co-processor, QCC5100
K7	PM_SLEEP_CLK3	VREG_1P8_SYS	DO	Sleep clock output to heart rate monitor sensor
XO and clock buffer circuits				
M1	XTAL_IN	–	AI	38.4 MHz crystal oscillator (XO) input
L1	XTAL_OUT	–	AO	38.4 MHz crystal oscillator (XO) output
K1	VREG_XO	–	AO	Linear regulator for XTAL circuits (internal use only)
K4, L3, L4, L6, M2, M4, M7, N2, N4, N5, N6	GND_XO	–	GNDP	Exclusive ground for XTAL circuits, which connects directly to the main ground plane through a single dedicated via; this ground pin should not be shared with any other ground pins.
M3	RF_CLK1	VREG_RF_CLK1	DO	RF (low-noise) XO clock buffer output (1 of 3)
M5	RF_CLK2	VREG_RF_CLK2	DO	RF (low-noise) XO clock buffer output (2 of 3)
M6	RF_CLK3	VREG_RF_CLK2	DO	RF (low-noise) XO clock buffer output (3 of 3)
K2	VREG_RF_CLK1	–	AO	Linear regulator for RF_CLK1 circuits (internal use only)
N1	VREG_RF_CLK2	–	AO	Linear regulator for RF_CLK2/RF_CLK3 circuits (internal use only)
L5	LN_BB_CLK1	VREG_L14A	DO	19.2 MHz, CXO input to QTI SoC
K5	LN_BB_CLK2	VREG_L14A	DO	19.2 MHz, input to WLAN
K3	LN_BB_CLK3	VREG_L14A	DO	19.2 MHz, input to NFC
K6	VDD_XO_RF	–	PI	Power supply for XO and RF clock buffer LDOs
PMIC power infrastructure				

Table 2-7 Pin descriptions – general housekeeping functions (cont.)

Pin #	Pin name and/or function	Pin characteristics [a]		Functional description
		Voltage	Type	
M17	REF_BYP	–	AO	Bypass capacitor for the dedicated master bandgap regulator; this reference must only be used for the master bandgap and must not be used as a general reference output.
M16	REF_GND	–	GNDP	Dedicated master bandgap ground reference connection; should be isolated from all other grounds – the pin should be connected directly to the main ground plane (with a via at the pad) and to the REF_BYP capacitor negative terminal.

[a] See Table 2-1 for pin voltage and type definitions.
[b] See Section 3.9 in *PMW5100 Data Sheet* (80-22756-1) for pull down resistor value.

Table 2-8 Pin descriptions – user interface functions

Pin #	Pin name	Pin type [a]	Functional description
Haptics			
E9	VSW_HAP_P	AO	Haptics H-bridge driver output plus (+)
F9	VSW_HAP_M	AO	Haptics H-bridge driver output minus (-)
E8, F8	VDD_HAP	PI	Haptics supply power input
E10, F10	PGND_HAP	GNDP	Haptics power ground

[a] See Table 2-1 for parameter and acronym definitions.

Table 2-9 Pin descriptions – IC-level interface functions

Pin #	Pin name and/or function	Pin characteristics [a]		Functional description
		Voltage	Type	
IC-level interfacing power supply				
J13	VIN_1P8_SYS	–	PI	Internal circuits I/O power supply in input. Connect to S4A
J12	VREG_1P8_SYS	–	PO	Internal circuits I/O power supply LDO output
L12	XVDD_BYP	–	AI	Internal XVDD LDO bypass capacitor input
Vref				
K13	VREF_MSM	–	AO	Reference for QTI SoC
Power on/off/reset control				
H6	KPD_PWR_N	XVDD	DI	Input pin generally connected to a keypad power-on button and when grounded, initiates the power-on sequence. Can also be configured for generating a stage 2 and/or stage 3 reset if held at a logic low for longer durations. Pulled up internally to 1.5 V via the XVDD regulator.
H12	PM_RESIN_N	VREG_1P8_SYS	DI	Input pin used for generating a stage 2 and/or stage 3 reset when held at a logic low.
H13	PM_PON_RESET_N	VREG_1P8_SYS	DO	Power-on reset output signal (active low), which is deasserted to take the QTI SoC out of reset after the PMIC power-on sequence has completed, and is asserted when a reset or shutdown commences.

Table 2-9 Pin descriptions – IC-level interface functions (cont.)

Pin #	Pin name and/or function	Pin characteristics [a]		Functional description
		Voltage	Type	
H11	PM_PS_HOLD	VREG_1P8_SYS	DI	Power supply hold control input. This signal's main purpose is to tell the PMIC device to keep its power supplies on, and it can initiate a reset or power-down when asserted low; this signal comes from the QTI SoC's PS_HOLD output.
System power management interface (SPMI) and I^2C signals				
J15	PM_SPMI_CLK	VREG_1P8_SYS	DI	SPMI communication bus clock signal
H15	PM_SPMI_DATA	VREG_1P8_SYS	DI, DO	SPMI communication bus data signal
J14	PM_I2C_CLK	VREG_1P8_SYS	DI	I2C communication clock signal
H14	PM_I2C_DATA	VREG_1P8_SYS	DI, DO	I2C communication data signal
Miscellaneous IC functions				
K12	TEST_EN_VPP	–	GNDC	Pin must be grounded externally
H10	FAULT_N	VREG_1P8_SYS	–	This signal will go low from high for PMIC internal fault. Add a test point in PCB for debug purpose only. Pulled up internally. Not to be connected/driving by any other circuit
L11	RID	–	AI	RID for factory mode detection.

a See Table 2-1 for pin voltage and type definitions.

Table 2-10 Pin descriptions – general-purpose input/output functions

Pin #	Pin name	Primary function	Pin type [a] [b]	Functional description for SW5100 [c]
Predefined GPIO functions – available only at the assigned GPIOs				
E18	PM_GPIO_01	DISPLAY_ELVDD_EN	LV	Display AVDD power supply enable output
E19	PM_GPIO_02	DISPLAY_RESET_N	LV	Display reset control output
E17	PM_GPIO_03	QCC_CLK_REQ	LV	QCC5100 RF clock request input and it enables RFCLK2
D18	PM_GPIO_04	NFC_CLK_REQ	LV	NFC RF clock request input (LN_BB_CLK3 output enabled)
D20	PM_GPIO_05	PM_I2C_IRQ	LV	PMIC I2C interrupt output signal
D19	PM_GPIO_06	QCC_PWR_DWN_N	LV	QCC5100 reset control output
D17	PM_GPIO_07	QCC_SW_CTRL	LV	S5A voltage and mode control input (from QCC5100)
F17	PM_GPIO_08	DEEP_SLEEP_EN	LV	SW5100-0/SW5100P-0 deep sleep enable output
F18	PM_GPIO_09	KYPD_UP_N	MV	Volume up keypad input
F19	PM_GPIO_10	KYPD_SIDE1_N	MV	Side button input This GPIO must have test point for debug purpose. Add a 0 ohm series resistor when used for other general purpose
G17	PM_GPIO_11	KYPD_SIDE2_N	MV	Side button input
G18	PM_GPIO_12	HRM_REQ	MV	HRM power rails transitions to NPM from LPM. If HRM_BOOST_EN is not used, PM_GPIO_12 should float in the design. It should not be re-purposed for any other general purpose.
G19	PM_GPIO_13	HAPTICS_TRIGGER (optional)	MV	Haptics system gets enabled and control the BoB output voltage

Table 2-10 Pin descriptions – general-purpose input/output functions (cont.)

Pin #	Pin name	Primary function	Pin type [a] [b]	Functional description for SW5100 [c]
G20	PM_GPIO_14	Reserved	MV	Do not use this GPIO for any other general purpose
H18	PM_GPIO_15	QCC5100 Boot mode selection	MV	Connect to GPIO_16 of QCC5100
H19	PM_GPIO_16	FORCED_USB_BOOT	MV	Factory mode detection output to control the SW5100 FORCED_USB_ BOOT mode GPIO

[a] See Table 2-1 for pin voltage and type definitions.
[b] For LV, GPIOs has only one voltage supply, L19 A.

For MV, GPIOs voltage supply is configurable. GPIOxx_DIG_VIN_CTL register has option to choose between VIN0 (VIN0 = VPH_PWR) and VIN1 (VIN1 = VREG_L19 A). When a MV GPIO is configured with VIN0 = VPH_PWR, the voltage at that GPIO must not exceed the instantaneous voltage at VPH_PWR. Similarly for LV GPIOs or MV GPIOs configured in VIN1 = VREG_L19A, the voltage must not exceed 1.8 V.

[c] All PMIC GPIOs have predefined functions and controlled in software or in PMIC OTP image. Do not re-purpose GPIOs for general purpose. Check with QTI if there are any special requirements.

Table 2-11 Pin description – audio

Pin #	Pin name and/or function	Pin name or alternate function	Pin type [a]	Functional description [b]
Audio codec – analog audio outputs and related interfaces				
A14, A16, B14, B16	VDD_SPKR	–	PI	Class-D speaker power amplifier supply
A17, B17, C16	SPKR_OUT_P	–	AO	Class-D speaker driver output, plus (+)
A15, B15, C15	SPKR_OUT_M	–	AO	Class-D speaker driver output, minus (-)
G15	SPKR_VSNS_P	–	AI	Speaker IV sense protection input plus (+)
F15	SPKR_VSNS_M	–	AI	Speaker IV sense protection input minus (-)
F12	SWR_CLK	–	DI	SoundWire clock input
G12	SWR_DATA	–	DI, DO	SoundWire data
C14, C17	GND_SPKR	–	GND	Class-D speaker power amplifier ground
G13	CDC_MIC_BIAS_1	–	AO	Microphone bias 1
F13	CDC_MIC_BIAS_2	–	AO	Microphone bias 2
A11	CDC_IN1_P	–	AI	Analog audio input 1 plus (+)
B11	CDC_IN1_M	–	AI	Analog audio input 1 minus (-)
A12	CDC_IN2_P	–	AI	Analog audio input 2 plus (+)
B12	CDC_IN2_M	–	AI	Analog audio input 2 minus (-)

[a] See Table 2-1 for pin voltage and type definitions.
[b] Codec resources (microphone bias outputs, analog audio inputs, and class-D speaker out) are controlled in software through SWR interface. SPMI bus has no access to codec registers, so it is mandatory to connect the SWR_CLK and SWR_DATA to SW5100/QCC5100 to use any of the codec resources in the design.

Table 2-12 Pin descriptions – common grounds

Pin #	Pin name	Pin type [a]	Functional description
Note: This table only includes common ground pins. Power ground pins are grouped with their respective modules, and can be found in the previous tables.			
A13, B13, C10, C11, C12, C13, C18, C19, C9, D10, D11, D12, D13, D14, D15, D16, D22, D6, D8, E11, E12, E13, E14, E15, E16, E21, E7, F11, F14, F16, F21, F5, F7, G10, G11, G14, G16, G7, G8, G9, H16, H5, H9, J20, K20, L21	CMN_GND	GNDC	

[a] See Table 2-1 for pin voltage and type definitions.

2.3 Pin assignments – top

A high-level view of the top pin assignments is shown in the following figure.

The text within the figure is difficult to read when viewing an 8½ inch × 11 inch hard copy.

Other viewing options are available and defined in Pin map – bottom.

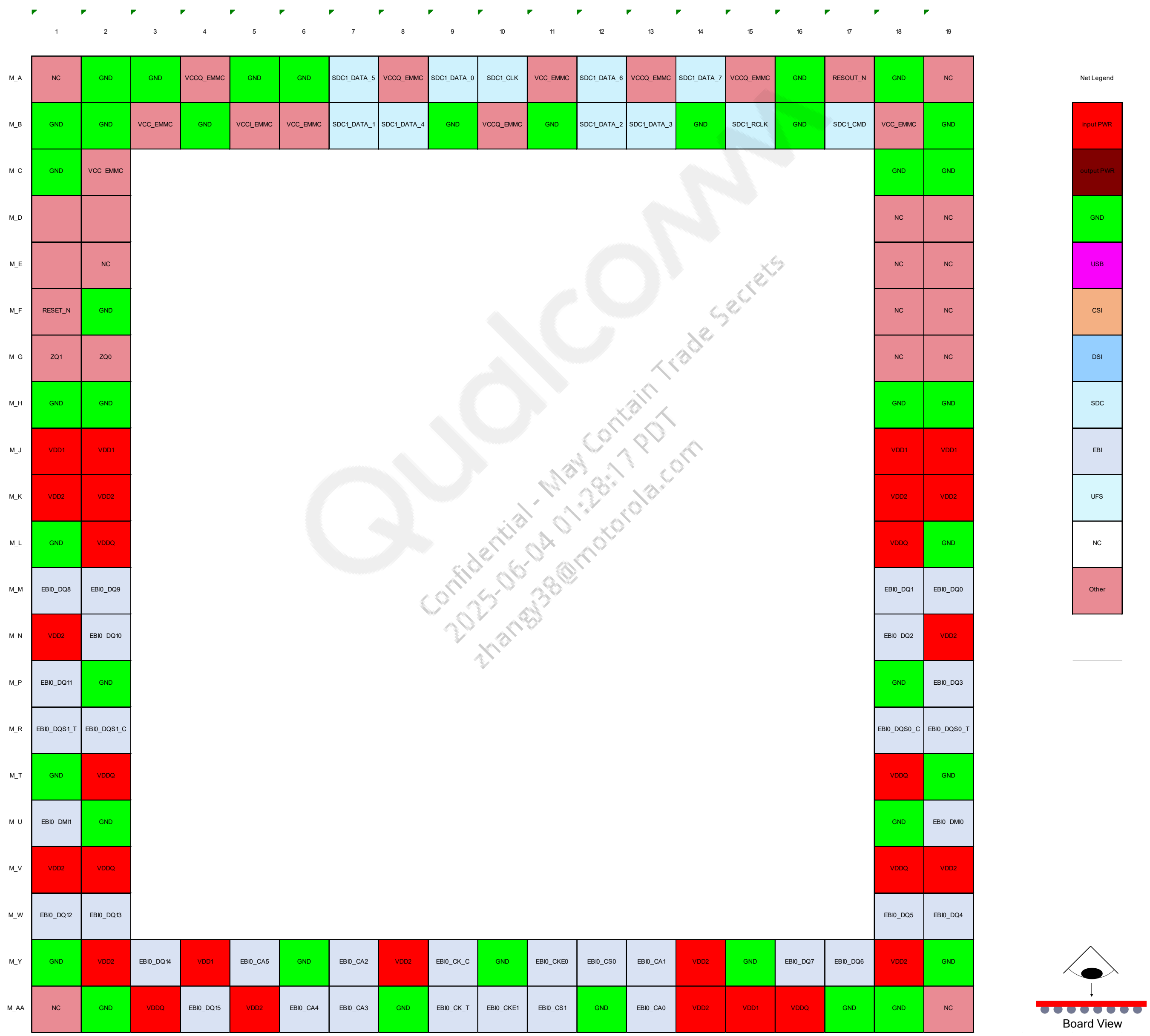

Figure 2-2 SW5100-0/SW5100P-0 top pin assignments

2.3.1 Pin descriptions – top

The top pins are described in the following tables.

Table 2-13 SW5100-0/SW5100P-0 top pin descriptions – general pins

Pin #	Pin name	Pin characteristics		Functional description
		Voltage	Type	
M_F1	RESET_N	P1	DO	LPDDR4x reset
M_AA13	EBI0_CA0	P1	DO	LPDDR4x command/address bit 0
M_Y13	EBI0_CA1	P1	DO	LPDDR4x command/address bit 1
M_Y7	EBI0_CA2	P1	DO	LPDDR4x command/address bit 2
M_AA7	EBI0_CA3	P1	DO	LPDDR4x command/address bit 3
M_AA6	EBI0_CA4	P1	DO	LPDDR4x command/address bit 4
M_Y5	EBI0_CA5	P1	DO	LPDDR4x command/address bit 5
M_Y11	EBI0_CKE0	P1	DO	LPDDR4x clock enable 0
M_AA10	EBI0_CKE1	P1	DO	LPDDR4x clock enable 1
M_Y9	EBI0_CK_C	P1	DO	LPDDR4x differential clock (negative)
M_AA9	EBI0_CK_T	P1	DO	LPDDR4x differential clock (positive)
M_Y12	EBI0_CS0	P1	DO	LPDDR4x Chip Select 0
M_AA11	EBI0_CS1	P1	DO	LPDDR4x Chip Select 1
M_U19	EBI0_DMI0	P1	DO	LPDDR4x data mask for byte 0
M_U1	EBI0_DMI1	P1	DO	LPDDR4x data mask for byte 1
M_M19	EBI0_DQ0	P1	B	LPDDR4x data bit 0
M_M18	EBI0_DQ1	P1	B	LPDDR4x data bit 1
M_N18	EBI0_DQ2	P1	B	LPDDR4x data bit 2
M_P19	EBI0_DQ3	P1	B	LPDDR4x data bit 3
M_W19	EBI0_DQ4	P1	B	LPDDR4x data bit 4
M_W18	EBI0_DQ5	P1	B	LPDDR4x data bit 5
M_Y17	EBI0_DQ6	P1	B	LPDDR4x data bit 6
M_Y16	EBI0_DQ7	P1	B	LPDDR4x data bit 7
M_M1	EBI0_DQ8	P1	B	LPDDR4x data bit 8
M_M2	EBI0_DQ9	P1	B	LPDDR4x data bit 9
M_N2	EBI0_DQ10	P1	B	LPDDR4x data bit 10
M_P1	EBI0_DQ11	P1	B	LPDDR4x data bit 11
M_W1	EBI0_DQ12	P1	B	LPDDR4x data bit 12
M_W2	EBI0_DQ13	P1	B	LPDDR4x data bit 13
M_Y3	EBI0_DQ14	P1	B	LPDDR4x data bit 14
M_AA4	EBI0_DQ15	P1	B	LPDDR4x data bit 15
M_R18	EBI0_DQS0_C	P1	B	LPDDR4x differential data strobe for byte 0 – negative
M_R19	EBI0_DQS0_T	P1	B	LPDDR4x differential data strobe for byte 0 – positive
M_R2	EBI0_DQS1_C	P1	B	LPDDR4x differential data strobe for byte 1 – negative

Table 2-13 SW5100-0/SW5100P-0 top pin descriptions – general pins (cont.)

Pin #	Pin name	Pin characteristics		Functional description
		Voltage	Type	
M_R1	EBI0_DQS1_T	P1	B	LPDDR4x differential data strobe for byte 1 – positive
M_A17	RESOUT_N	P3	DO	Reset Output
M_A10	SDC1_CLK	P3	B-NP:pdpukp	eMMC Clock
M_B17	SDC1_CMD	P3	B-PD:nppukp	eMMC command SDC1
M_A9	SDC1_DATA_0	P3	B-PD:nppukp	SDC1 – eMMC data bit 0
M_B7	SDC1_DATA_1	P3	B-PD:nppukp	SDC1 – eMMC data bit 1
M_B12	SDC1_DATA_2	P3	B-PD:nppukp	SDC1 – eMMC data bit 2
M_B13	SDC1_DATA_3	P3	B-PD:nppukp	SDC1 – eMMC data bit 3
M_B8	SDC1_DATA_4	P3	B-PD:nppukp	SDC1 – eMMC data bit 4
M_A7	SDC1_DATA_5	P3	B-PD:nppukp	SDC1 – eMMC data bit 5
M_A12	SDC1_DATA_6	P3	B-PD:nppukp	SDC1 – eMMC data bit 6
M_A14	SDC1_DATA_7	P3	B-PD:nppukp	SDC1 – eMMC data bit 7
M_B15	SDC1_RCLK	P3	DI-PD:pdpukp	eMMC return Clock -SDC1
M_G2	ZQ0	–	AI	Pin for ZQ0 calibration
M_G1	ZQ1	–	AI	Pin for ZQ1 calibration

Table 2-14 SW5100-0/SW5100P-0 top pin descriptions – ground, power pins

Pin #	Pin name	Functional description
M_A1, M_A19, M_AA1, M_AA19, M_D18, M_D19, M_ E18, M_E19, M_E2, M_F18, M_F19, M_G18, M_G19	NC	No connection
M_B5	VCCI_EMMC	Power for eMMC – core IO
M_A13, M_A15, M_A4, M_A8, M_B10	VCCQ_EMMC	Power for memory I/O
M_A11, M_B18, M_B3, M_B6, M_C2	VCC_EMMC	Power for eMMC – control
M_A16, M_A18, M_A2, M_A3, M_A5, M_A6, M_AA12, M_AA17, M_AA18, M_AA2, M_AA8, M_B1, M_B11, M_B14, M_B16, M_B19, M_B2, M_B4, M_B9, M_C1, M_C18, M_C19, M_F2, M_H1, M_H18, M_H19, M_H2, M_L1, M_L19, M_P18, M_P2, M_T1,M_T19, M_U18, M_U2, M_Y1, M_Y10, M_Y15, M_Y19, M_Y6	GND	Ground
M_AA15, M_J1, M_J18, M_J19, M_J2, M_Y4	VDD1	Power for DDR memory core – 1.8 V
M_AA14, M_AA5, M_K1, M_K18, M_K19, M_K2, M_N1, M_N19, M_V1, M_V19, M_Y14, M_Y18, M_Y2, M_Y8	VDD2	Power for DDR memory core – 1.2 V
M_AA16, M_AA3, M_L18, M_L2, M_T18, M_T2, M_V18, M_V2	VDDQ	DDR IO voltage supply – 0.6 V
M_D1	VSF1	Vendor specific function 1
M_D2	VSF3	Vendor specific function 3
M_E1	VSF2	Vendor specific function 2

3 Electrical specifications

For electrical specifications of SW5100-0/SW5100P-0 chipset, see *SW5100-1/SW5100P-1 Data Sheet* (80-24634-1) document.

3.1 Power delivery network (PDN) specifications

For more information on the PDN specifications, see *SW5100 Power Delivery Network Specifications* (80-24634-1P) document.

4 Mechanical information

4.1 Device physical dimensions

The SW5100-0/SW5100P-0 is available in the MPSP 688 that includes dedicated ground pins for improved grounding, mechanical strength, and thermal continuity. The MPSP 688 has a

10.4 mm by 8.6 mm body, with a maximum height of 0.48 mm. Pin A1 is located by an indicator mark on the top of the package, and by the ball pattern when viewed from below. A simplified version of the MPSP 688 outline drawing is shown in the following figure.

Click the following link to download the *Package Outline Drawing, MPSP688, 10.4 mm × 8.6 mm × 0.48 mm, ST94, M102, SB118, PB 144 NSP, PL1, MEP* (NT90-PU628-1) from the CreatePoint website.

https://createpoint.qti.qualcomm.com/search/contentdocument/stream/dcn/NT90-PU628-1

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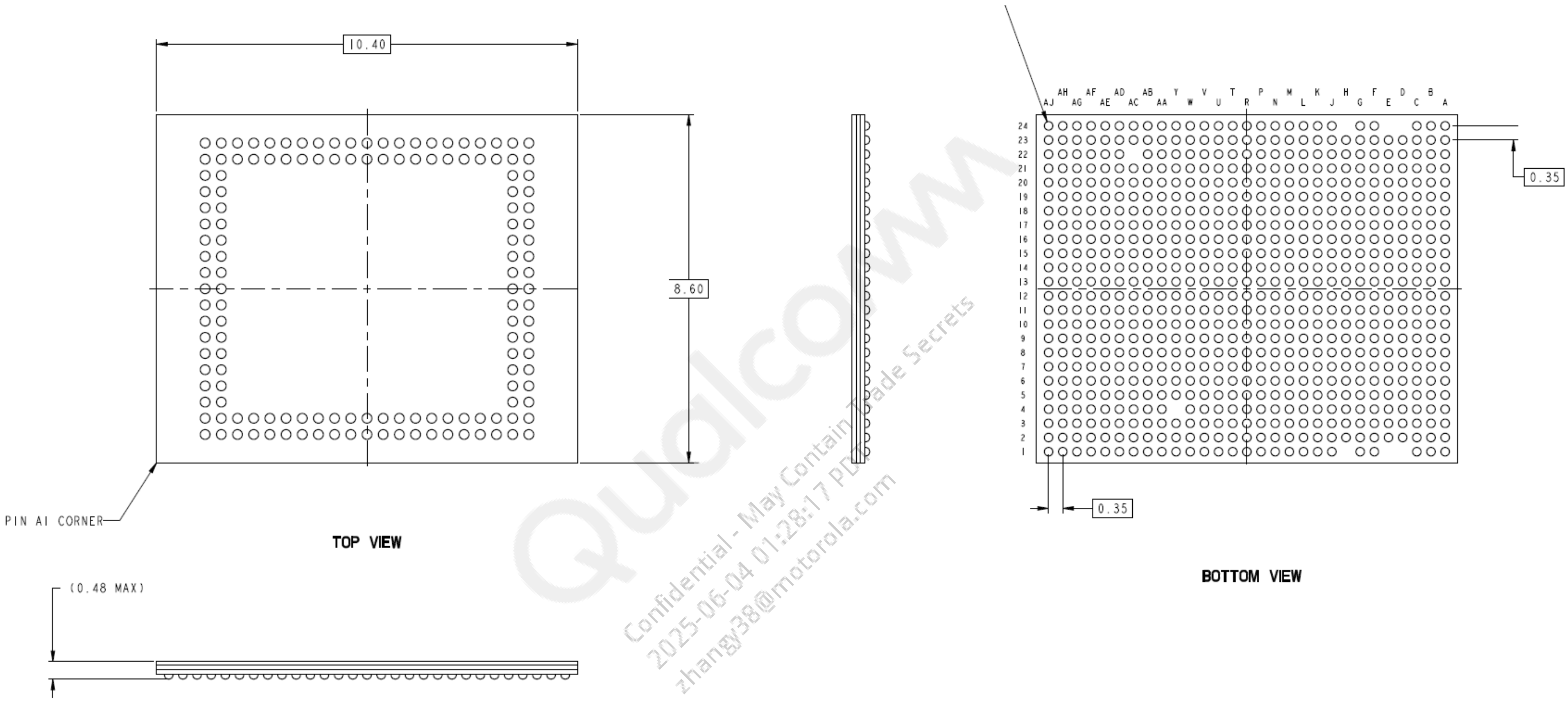


Figure 4-1 MPSP 688 outline drawing

NOTE This is a simplified outline drawing. Click the link on the previous page to download the complete, up-to-date package outline drawing.

4.2 Part marking

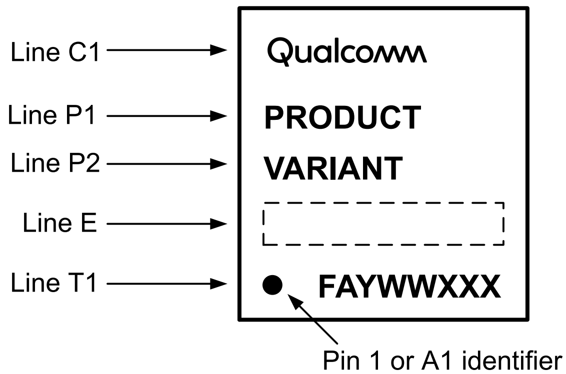


Figure 4-2 Device marking (top view, not to scale)

Table 4-1 Device marking line definitions

Line	Marking	Description
C1	Qualcomm	Qualcomm name
P1	PRODUCT	Qualcomm Technologies, Inc. (QTI) product name ▪ SW5100 ▪ SW5100P
P2	VARIANT	Device variant information ▪ See Table 4-4 for the assigned values.
E	Blank or random	Additional content, as necessary
T1	•	Pin 1 or pin A1 indicator
	FAYWWXXX	F = supply source code ▪ F = J (Samsung) ▪ F = P (SMIC) A = assembly site code ▪ A = C (Amkor, Korea) ▪ A = X (Shinko, Japan) Y = single/last digit of year WW = two-digit work week of year specified by Y XXX = traceability number

For complete marking definitions of all SW5100-0/SW5100P-0 variants and revisions, see the *SW5100-0/SW5100P-0 Device Revision Guide* (80-24633-4) document.

Table 4-2 QFPROM_CORR_PTE_ROW0_LSB (0x1B44180)

Bit location	Name	Description
bits [27:20]	FEATURE_ID	These bits are used for defining various feature variants (see Table 4-4).
bits [19:0]	JTAG_ID	These bits map to bits of the hardware revision number (see Table 4-4).

4.3 Device ordering information

The Oracle short description is used to order QTI products, and is present on both the customer label and this document. The short description includes the product name, configuration code, package type, product revision code, source code, and feature code/program ID of the part.

This device can be ordered using the identification code shown in the following table.

Table 4-3 Device identification code

Device ID code	AAA-AAAA	-P	-TTTTTT	NNNN	A	+FF	-EE	-RR	-S	-BB or -PID [a]
Symbol definition	**Product name**	**Configuration code**	**Package type**	**Number of pins**	**Package variable**	**Additional package information**	**Shipping package**	**Product revision**	**Source code**	**Feature code**
Example 1	SW-5100	-0	-MPSP	688			-TR	-00	-0	-AA
Example 2	SW-5100P	-0	-MPSP	688			-TR	-00	-0	-AA

[a] The feature code (BB) and the program ID (PID) are mutually exclusive. A product may have one of them or none of them, but it will never have both. If there is no feature code/program ID, this field is blank, and the Oracle short description ends after the source configuration code (S).

For example:

- Example 1: SW-5100-0-MPSP688-TR-00-0-AA
- Example 2: SW-5100P-0-MPSP688-TR-00-0-AA

NOTE The shipping package is either TR (tape and reel) or MT (matrix tray).

Device identification details for all samples available to date are summarized in the following table.

Table 4-4 Device identification details

Device	Sample type	Variant (PRR-BB) P = product configuration code RR = product revision code BB = feature code [a]	FEATURE_ID (see Table 4-3) [b]	Hardware revision number (JTAG_ID – see Table 4-3)	Hardware version	Source configuration code (S) [c]	Comments	Sample date
SW5100	EA	000-AA	0x1	0x0 01A2 0E1	v1.0	0	SW5100 (000-0-AA), MPSP688, 4 x A53, MEP configuration with modem	–
SW5100P	EA	000-AA	0x1	0x0 01A5 0E1	v1.0	0	SW5100P (000-0-AA), MPSP688, 4 x A53, MEP configuration with no modem	–
SW5100	ES	000-AA	0x1	0x0 01A2 0E1	v1.0	0	SW5100 (000-0-AA), MPSP688, 4 x A53, MEP configuration with modem	09/03/2021
SW5100P	ES	000-AA	0x1	0x0 01A5 0E1	v1.0	0	SW5100P (000-0-AA), MPSP688, 4 x A53, MEP configuration with no modem	09/03/2021
SW5100	ES	001-AA	0x1	0x0 01A2 0E1	v2.0 [d]	0	SW5100 (001-0-AA), MPSP688, 4 x A53, MEP configuration with modem	03/03/2022
SW5100P	ES	001-AA	0x1	0x0 01A5 0E1	v2.0 [d]	0	SW5100P (001-0-AA), MPSP688, 4 x A53, MEP configuration with no modem	03/09/2022
SW5100	CS	001-AA CS date code is as follows: ▪ Amkor: 211 ▪ Shinko: 203	0x1	0x0 01A2 0E1	v2.0 [d]	0	SW5100 (001-0-AA), MPSP688, 4 x A53, MEP configuration with modem	05/20/2022
SW5100P	CS	001-AA CS date code is as follows: ▪ Amkor: 211	0x1	0x0 01A5 0E1	v2.0 [d]	0	SW5100P (001-0-AA), MPSP688, 4 x A53, MEP configuration with no modem	05/20/2022

Table 4-4 Device identification details (cont.)

Device	Sample type	Variant (PRR-BB) P = product configuration code RR = product revision code BB = feature code [a]	FEATURE_ID (see Table 4-3) [b]	Hardware revision number (JTAG_ID – see Table 4-3)	Hardware version	Source configuration code (S) [c]	Comments	Sample date
		▪ Shinko: 203						

[a] BB is the feature code that identifies an IC's specific feature set, which distinguishes it from other versions or variants. Feature sets are detailed in the Comments column.
[b] The FEATURE_ID combined with the hardware revision number (JTAG_ID) defines unique product variants. This information is shown for situations where other device identification information (such as device marking information) is not easily accessible.
[c] S is the source configuration code that identifies all of the qualified die fabrication-source combinations available when the particular sample type was shipped. S values are defined in Table 4-5.
[d] The SoC portion remains unchanged between RR = 00 and RR = 01. Only the PMIC portion is updated.

Table 4-5 Source configuration code

S value	F value = J	F value = P	A value = C	A value = X
0	Samsung	SMIC	Amkor, Korea	Shinko, Japan
Other columns and rows will be added in future revisions of this document, if needed.				

4.3.1 Daisy chain devices

For daisy chain part information, contact the Qualcomm Sales team for support.

4.4 Device moisture sensitivity level

Plastic-encapsulated surface mount packages are susceptible to damage induced by absorbed moisture and high temperature. A package's moisture sensitivity level (MSL) indicates its ability to withstand exposure after it is removed from its shipment bag, while it is on the factory floor awaiting PCB installation. A low MSL rating is better than a high rating; a low MSL device can be exposed on the factory floor longer than a high MSL device. All pertinent MSL ratings are summarized in the following table.

Table 4-6 MSL ratings summary

MSL	Out-of-bag floor life	Comments
1	Unlimited	≤ 30°C/85% RH
2	1 year	≤ 30°C/60% RH
2a	4 weeks	≤ 30°C/60% RH
3	168 hours	≤ 30°C/60% RH; **SW5100-0/SW5100P-0 rating**
4	72 hours	≤ 30°C/60% RH
5	48 hours	≤ 30°C/60% RH
5a	24 hours	≤ 30°C/60% RH
6	Mandatory baking before use. After baking, must be reflowed within the time limit specified on the label.	≤ 30°C/60% RH

QTI follows the latest IPC/JEDEC J-STD-020 standard revision for moisture-sensitivity qualification. ***The SW5100-0/SW5100P-0 devices are classified as MSL3; the qualification temperature was 255ºC.*** This qualification temperature (255°C) should not be confused with the peak temperature within the recommended solder reflow profile.

4.5 Thermal characteristics

Rather than providing thermal resistance values Θ_{JC} and Θ_{JA}, validated thermal package models are provided through the CreatePoint website. Designers can extract thermal resistance values by conducting their own thermal simulations.

NOTE Click the following link to download the *SW5100-0 Package Thermal Model Icepak* (HS11-24633-5HW) or *SW5100-0 Package Thermal Model FloTHERM* (HS11-24633-6HW) from the CreatePoint website.

Thermal Package Model Icepak

https://createpoint.qti.qualcomm.com/search/contentdocument/stream/dcn/HS11-24633-5HW

Or

Thermal Package Model FloTHERM

https://createpoint.qti.qualcomm.com/search/contentdocument/stream/dcn/HS11-24633-6HW

After successfully logging in, the document is downloaded.

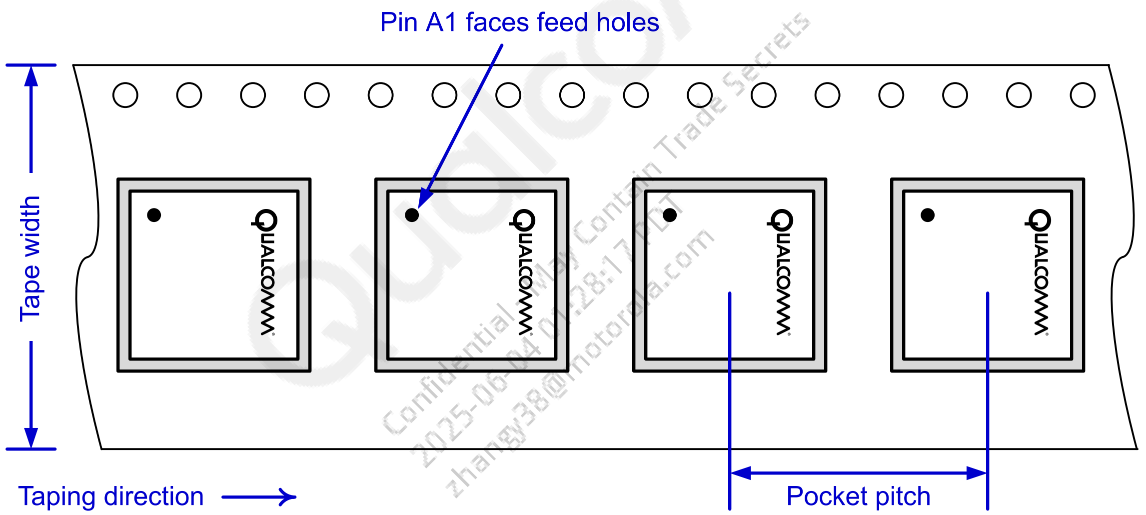
5 Carrier, storage, and handling information

5.1 Carrier

5.1.1 Tape and reel information

All QTI tape carrier systems conform to EIA-481 standards.

A simplified sketch of the SW5100-0/SW5100P-0 tape carrier is shown in the following figure, including the proper part orientation, maximum number of devices per reel, and key dimensions.



Tape feed:	Single	Reel diameter:	330 mm	Tape width:	24 mm
Units per reel:	4000	Hub diameter:	178 mm	Pocket pitch:	12 mm

Figure 5-1 Carrier tape drawing with part orientation

Tape-handling recommendations are shown in the figure below:

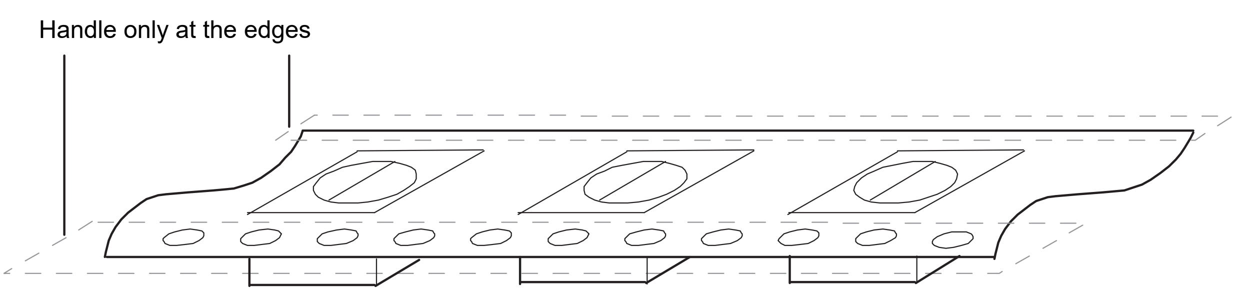


Figure 5-2 Tape handling

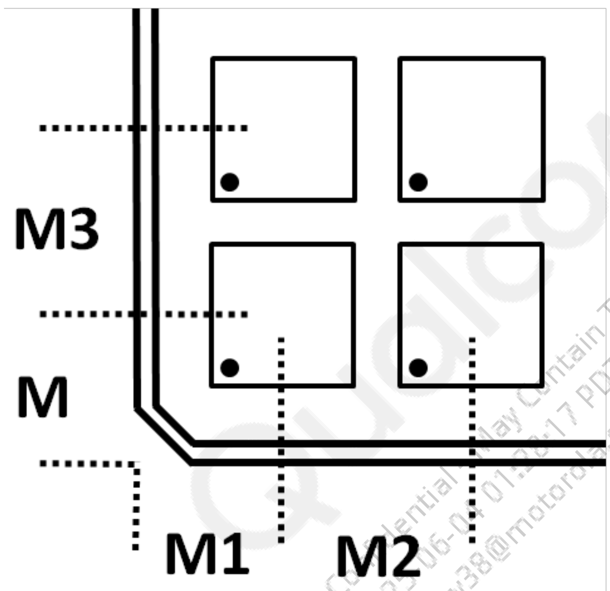
5.1.2 Matrix tray information

All QTI matrix tray carriers confirm to JEDEC standards.

The device pin 1 is oriented to the chamfered corner of the matrix tray.

Each tray of the SW5100-0/SW5100P-0 contains up to 210 devices. Production orders of the SW5100-0/SW5100P-0 that are shipped in matrix tray carriers will be in [10 + 1] tray stacks of [2100] units. The stacking configuration and quantity for sample orders will vary.

See the following figure for matrix-tray key attributes and dimensions.



Key dimensions	
Array	10 x 21 (210)
M	10.35 mm
M1	11.00 mm
M2	14.65 mm
M3	12.80 mm

Figure 5-3 Matrix-tray key attributes and dimensions

5.2 Storage

5.2.1 Bagged storage conditions

SW5100-0/SW5100P-0 devices delivered in tape and reel carriers must be stored in sealed, moisture barrier, anti-static bags. See *IC Products Packing Method* (80-VK055-1) for the expected shelf life.

5.2.2 Out-of-bag duration

The out-of-bag duration is the time a device can be on the factory floor before being installed onto a PCB.

5.3 Handling

Tape handling was described in Tape and reel information. Other (IC-specific) handling guidelines are presented in the following subsections.

5.3.1 Baking

It is **not necessary** to bake the SW5100-0/SW5100P-0 if the conditions specified in Bagged storage conditions and Out-of-bag duration have **not been exceeded**.

It is **necessary** to bake the SW5100-0/SW5100P-0 if any condition specified in Bagged storage conditions or Out-of-bag duration has **been exceeded**. The baking conditions are specified on the moisture-sensitive caution label attached to each bag; see the *IC Products Packing Method* (80-VK055-1) document for details.

CAUTION: If baking is required, the devices must be transferred into trays that can be baked to at least 125°C. Devices should not be baked in tape and reel carriers at any temperature.

5.3.2 Electrostatic discharge

Electrostatic discharge (ESD) occurs naturally in laboratory and factory environments. An established high-voltage potential is always at risk of discharging to a lower potential. If this discharge path is through a semiconductor device, destructive damage may result.

ESD countermeasures and handling methods must be developed and used to control the factory environment at each manufacturing site.

QTI products must be handled according to the ESD Association standard: ANSI/ESD S20.20-1999, *Protection of Electrical and Electronic Parts, Assemblies, and Equipment*.

5.4 Bar code label and packing for shipment

See the *IC Products Packing Method* (80-VK055-1) document for all packing-related information, including bar code label details.

6 PCB mounting guidelines

6.1 RoHS compliance

The device complies with the requirements of the EU RoHS directive. Its SnAgCu solder balls use SAC125Ni composition. A product material declaration (PMD) that provides RoHS and other product environmental governance information is published when the data is available.

6.2 SMT assembly guidelines

For recommendations on SMT process development, see the *SMT Assembly Guidelines* (SM80-P0982-1).

NOTE Click the following link to download the *SMT Assembly Guidelines* (SM80-P0982-1) from the CreatePoint website.

https://createpoint.qti.qualcomm.com/search/contentdocument/stream/dcn/SM80-P0982-1

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7 Part reliability

7.1 Reliability evaluation summary

SW5100 reliability evaluation report for MEP MPSP688 device, foundry source SEC and SMIC.

Table 7-1 Silicon reliability results

Tests, standards and conditions	Sample size	Result
ELFR in DPPM HTOL: JESD22-A108-A Total samples from more than three different wafer lots	2186 [a]	Pass DPPM < 1000
HTOL in FIT (l) failure in billion device hours HTOL: JESD22-A108-A Total samples from more than three different wafer lots	80	Pass FIT < 100
Mean time to failure (MTTF) t = 1/l in million hours (Total samples from more than three different wafer lots)	80	Pass > 10
ESD – Human-body model (HBM) rating JESD22-A114-F Target 1000 V Total samples from one wafer lot.	21	Pass ± 1000 V
ESD – Charge-device model (CDM) rating JESD22-C101-D Target 250 V Total samples from one wafer lot.	3	Pass ± 250 V
Latch-up (I-test): EIA/JESD78C Trigger current: ±100 mA; temperature: 85°C Total samples from one wafer lot.	3	Pass
Latch-up (Vsupply overvoltage): EIA/JESD78C Trigger voltage: stress at 1.5 × Vdd max per device specification; temperature: 85°C Total samples from one wafer lot	3	Pass

[a] Qualification data is leveraged from the same process technology node.

Table 7-2 Package reliability results

Tests, standards and conditions	ATK Sample size	Shinko Sample size	Result
Moisture resistance test (MRT): J-STD-020-C Reflow at 260 +0/-5 °C, MSL3 Total samples from three different assembly lots	552	552	Pass
Temperature cycle: JESD22-A104-D Temperature: -55°C to 125°C; number of cycles: 1000	231	231	Pass

Table 7-2 Package reliability results (cont.)

Tests, standards and conditions	ATK Sample size	Shinko Sample size	Result
Soak time at minimum/maximum temperature: 8-10 minutes Cycle rate: 2 cycles per hour (CPH) **Preconditioning:** JESD22-A113-F MSL 3, reflow temperature: 260C +0/-5°C Total samples from three different assembly lots			
Unbiased highly accelerated stress test: JESD22-A118 130°C / 85% RH and 96 hours duration Preconditioning: JESD22-A113-F MSL 3, reflow temperature: 260C +0/-5°C Total samples from three different assembly lots	231	231	Pass
Biased highly accelerated stress test: JESD22-A110 130°C /85% RH and 96 hour duration or 110°C /85% RH and 264-hours duration Preconditioning: JESD22-A113-F MSL 3, reflow temperature: 260°C +0/-5°C	90	90	Pass
High-Temperature Storage Life: JESD22-A103-C Temperature 150°C, 500, 1000 hours Total samples from three different assembly lots	231	231	Pass
Flammability UL-STD-94 Flammability test – Not required UL-STD-94 Qualcomm's ICs are exempt from the flammability requirements due to their sizes per UL/EN 60950-1, as long as they are mounted on materials rated V-1 or better. Most PWBs onto which our ICs mounted are rated V-0 (better than V-1)	N/A	N/A	N/A
Physical dimensions: JESD22-B100-B Case outline drawing: Qualcomm internal document Total samples from three different assembly lots	30	30	Pass
Solder ball shear: JESD22-B117A Total samples from three different assembly lots	15	15	Pass

7.2 Qualification sample description

Table 7-3 Device characteristics

Category	Definition
Device name	SW5100-0/SW5100P-0
Package type	MPSP 688
Package body size	10.4 mm × 8.6 mm × 0.48 mm
Ball count	688
Ball composition	SAC125/Ni
Fab process	4 nm-LPX
Supply sources (fab sites)	▪ Samsung ▪ SMIC
Assembly sites	Amkor, Korea
Solder ball pitch	0.4 mm

8 Revision history

Bars appearing in the margin (as shown here) indicate where technical changes have occurred for this revision. The following table lists the technical content changes for all revisions.

Revision	Date	Description
AA	April 20, 2021	Initial release
AB	April 30, 2021	▪ Figure 1-1 *SW5100-0/SW5100P-0 functional block diagram and example application*: Updated the block diagram ▪ Table 2-1 *I/O description (pad type) parameters*: Added GNDC and GNDP pad descriptions ▪ Section 2.2.1 *Pin map – bottom*: Added CreatePoint link to download the Pin Assignment and GPIO Spreadsheet document ▪ Figure 2-1 *SW5100-0/SW5100P-0 bottom pin assignments* and Figure 2-2 *SW5100-0/SW5100P-0 top pin assignments*: Updated the pin map ▪ Table 2-3 *SW5100-0/SW5100P-0 bottom pin descriptions – general- purpose input/output ports*: Added LPI GPIO descriptions
AC	June 2021	▪ Cover page: Updated RF related content and the outline diagram ▪ Figure 2-1 *SW5100-0/SW5100P-0 bottom pin assignments*: Updated the pin map diagram ▪ Table 2-2 *SW5100-0/SW5100P-0 bottom pin descriptions – general pins*: Updated the table ▪ Table 2-7 *Pin descriptions – general housekeeping functions*: Updated the table ▪ Table 2-9 *Pin descriptions – IC-level interface functions*: Updated the table ▪ Table 2-10 *Pin descriptions – general-purpose input/output functions*: Updated the table ▪ Figure 4-1 *MPSP 688 outline drawing*: Updated the figure ▪ Section 4.2 *Part marking* and Section 4.3 *Device ordering information*: Added these sections
AD	September 2021	▪ Cover page: Updated key features ▪ Figure 1-1 *SW5100-0/SW5100P-0 functional block diagram and example application*: Updated the block diagram ▪ Table 1-1 *Comparison of SW5100-0 and SW5100-1 chipsets*: Updated the table ▪ Table 2-1 *I/O description (pad type) parameters*: Updated P0 description ▪ Figure 2-1 SW5100-0/SW5100P-0 bottom pin assignments and Figure 2-2 SW5100-0/ SW5100P-0 top pin assignments: Updated the pin map ▪ Table 2-3 SW5100-0/SW5100P-0 bottom pin descriptions – general- purpose input/output ports: Updated LPI GPIO pads ▪ Table 2-7 Pin descriptions – general housekeeping functions: Updated functional descriptions ▪ Section 3.1 Power delivery network (PDN) specifications: Added a note ▪ Section 4.2 Part marking: Added JTAG register details ▪ Section 4.3 Device ordering information: Added ES related details ▪ Added the following sections: ▪ Section 4.4 Device moisture sensitivity level ▪ Section 4.5 Thermal characteristics ▪ Added the following chapters: ▪ Chapter 5 Carrier, storage, and handling information ▪ Chapter 6 PCB mounting guidelines
AE	February 2022	▪ Cover page: Updated the device description and key features section ▪ Figure 1-1 *SW5100-0/SW5100P-0 functional block diagram and example application*: Updated the block diagram

Revision	Date	Description
		■ Figure 2-1 *SW5100-0/SW5100P-0 bottom pin assignments and Figure 2-2 SW5100-0/ SW5100P-0 top pin assignments*: Updated the pin map ■ Updated the following tables: ■ Table 2-5 *Pin descriptions – input power management functions* ■ Table 2-7 *Pin descriptions – general housekeeping functions* ■ Table 2-10 *Pin descriptions – general-purpose input/output functions* ■ Table 2-11 *Pin description – audio* ■ Section 2.2.3 *Pin descriptions – bottom (PMIC)*: Removed no connect pins table
AF	March 4, 2022	■ Table 4-4 *Device identification details*: Added a new SW5100 ES variant. ■ Table 4-5 *Source configuration code*: Added assembly site details
AG	March 10, 2022	■ Table 4-4 *Device identification details*: Added a new SW5100P ES variant.
AH	May 2022	■ Cover page: Updated GPU frequency ■ Table 1-1 *Comparison of SW5100-0 and SW5100-1 chipsets*: Updated Bluetooth version of SW5100-1 and WLAN frequency of SW5100-1 ■ Chapter 3 *Electrical specifications*: Updated this topic ■ Table 4-1 *Device marking line definitions*: Updated Assembly site code ■ Table 4-4 *Device identification details*: Added CS related details ■ Table 4-5 *Source configuration code*: Updated Assembly site code ■ Section 7.1 *Reliability evaluation summary*: Added this topic ■ Section 7.2 *Qualification sample description*: Added this topic
Revision I was omitted in accordance with QTI document conventions.		
AJ	July 2022	■ Table 1-1 *Comparison of SW5100-0 and SW5100-1 chipsets*: Updated this table ■ Section 4.5 *Thermal characteristics*: Updated this topic
AK	September 2022	Added new assembly site details in: ■ Table 4-1 *Device marking line definitions* ■ Table 4-4 *Device identification details* ■ Table 4-5 *Source configuration code*
AL	October 2023	Figure 1-1 *SW5100-0/SW5100P-0 functional block diagram and example application*: Updated the block diagram
AM	December 2023	■ Table 4-3 *Device identification code*: Updated shipping package ■ Section 4.3 *Device ordering information*: Added a note on shipping package

For additional information or to submit technical questions, go to https://www.qualcomm.com/support

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